Engineered Intermediate-Mass Black Hole Systems: Infrastructure Constraints, Observable Residue, and a Multi-Messenger Adjudication Framework

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v2.8 (last revised 2026-10-24) — fifth paper of the set; companion to *The Macro Transcension Hypothesis* (Paper A), the inward-migration review (Paper B), the ω Cen campaign (Paper C), the migration economics (Paper D), the joint accretion bound (Paper F),
the residual X-ray census (Paper G), and the mass-tension paper (Paper H);
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Abstract

The framework developed here cannot adjudicate an engineered black-hole system ($H_{\rm eng}$) on any electromagnetic null. Its mid-infrared channel is one-sided, so its evidence saturates at $\ln f_d = -0.69$ however deep the photometry goes, and without a detection it can never favor $H_{\rm eng}$; the cap depends on the dormancy prior ($\ln f_d = -2.3$ at $f_d = 0.1$).

Rapidly spinning intermediate-mass black holes (IMBHs) in dense, old clusters are proposed destinations for computation-optimizing civilizations. We ask whether infrastructure can persist around a $2 \times 10^4\,M_\odot$ Kerr IMBH in the ω Cen core. A flyby Monte Carlo gives an allowed envelope from $\sim 10^2$ gravitational radii to the stripping radius ($\sim 4\times10^3$ AU). Unbound flybys never set the boundary; bound-remnant diffusion does, with a floor of a few 10^7 yr when a remnant is present. With outflow suppression engineered away, ambient fueling supports up to $7.6\times10^5\,L_\odot$ after a thin-disc spin-up and $\approx 10^3$–$4\times10^3\,L_\odot$ after a magnetically arrested (MAD) spin-up, which also leaves $a_\star \approx 0.035$–0.07, within 1σ of the natural-capture null and below the T5 trigger $a_\star < 0.1$. Bayes factors against an explicit null menu score three forward-modeled observables: a waste-heat floor, MAD regulation, and EMRI dephasing. On current data and forecast JWST depths the mid-infrared channel returns $\ln K = -0.47$ and the engineered-mass channel -0.03, a total of -0.50. It registers the 1967 pulsar discovery as a strong anomaly ($\ln K = +7.9$), collapses it once the resolving hypothesis joins the menu, and gives no false positive on two mundane cases; one pre-registered criterion fails and is reported. Results are conditional on the optimization premise and on unrefereed companion-paper inputs.

Keywords: Fermi paradox; SETI; technosignatures; intermediate-mass black holes; Omega Centauri; megastructures; magnetically arrested disks; Bayesian model selection; multi-messenger astronomy

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1 Introduction

1.1 The missing quadrant

Four papers in this series have examined the hypothesis that computation-optimizing civilizations migrate toward rapidly spinning massive black holes in dense old stellar systems. Paper A (Swanson, 2026c) argued the motive: the thermodynamic advantages of Kerr accretion efficiency, horizon entropy disposal, and Bekenstein-bounded information storage. Paper B (Swanson, 2026b) surveyed the candidates: the six-member family of inward-migration proposals and their falsifiability grades. Paper C (Swanson, 2026d) planned the search: eight instrument-matched programs targeting Omega Centauri (NGC 5139), the nearest strong IMBH candidate. Paper D (Swanson, 2026a) priced the move: closed-form crossover conditions between migration, densification, and seeding strategies.

Two questions remain unaddressed, in this series and, so far as we can determine, in the literature. First: could the infrastructure such a civilization requires actually persist at the destination? The destination is a demanding one. A cluster-core IMBH sits in a stellar environment three to four orders of magnitude denser than the solar neighborhood, threaded by gravitationally focused stellar traffic, and (if fueled) surrounded by a relativistic accretion flow. Black-hole computing proposals (Inoue and Yokoo, 2011; Opatrný et al., 2017; Dvali and Osmanov, 2023) treat the hole as an idealized resource and the environment as empty. Megastructure stability analyses (Wright, 2020; Raval and Srikanth, 2024; McInnes, 2026) treat single stars in isolation. The intersection, engineered structures bound to a Kerr hole inside a live cluster core, is unexamined.

Second: if the campaign of Paper C ever records an anomaly, by what procedure would the community decide what it had seen? Radio SETI has mature per-signal verification practice (Sheikh et al., 2021; Li et al., 2026) and Bayesian population inference (Grimaldi and Marcy, 2018), and multi-messenger astrophysics has quantitative coincidence formalisms (Ashton et al., 2018; Breschi et al., 2024), but no published framework adjudicates a candidate technosignature across multiple messengers against an explicit menu of astrophysical alternatives. Paper C's rule that "every anomaly is adjudicated by at least two independent messengers" was stated as policy; here we give it mathematical content.

The two questions are coupled, and that coupling is the reason they share a paper. A feasibility envelope is a prior: it tells the adjudicator where in parameter space an engineered system could sit, and therefore which anomalies deserve elevated scrutiny and which are excluded on engineering grounds regardless of how anomalous they appear. Forward-modeled residue channels are likelihoods: they specify what an engineered system would look like, channel by channel, so that Bayes factors can be computed rather than gestured at. The paper thus runs in one direction: constraints (§2) produce observables (§3) which feed an adjudication engine (§4) that we exercise on real data (§5).

1.2 Conditionality and register

Epistemic status: Everything downstream of §2 is conditional on the optimization premise examined critically in Papers A and B: that some long-lived technological lineages maximize discounted computation and act on the thermodynamic gradients described there. This paper does not defend that premise; it asks what follows from it observationally, and how a null or a detection would be established. The adjudication framework of §4 is independent of the premise and applicable to any technosignature search.

We work throughout with the fiducial system of Papers A and C: a black hole of mass $M = 2 \times 10^4\, M_\odot$ (bracketed by the $8.2 \times 10^3\, M_\odot$ kinematic lower bound of Häberle et al. 2024b and the $\sim 5 \times 10^4\, M_\odot$ N-body preferred value of González Prieto et al. 2025), spin left free, embedded in the ω Cen core: central density $\rho_0 \simeq 3 \times 10^3\, M_\odot\, \mathrm{pc}^{-3}$, core radius $r_c \simeq 3.6$ pc, line-of-sight velocity dispersion $\sigma \simeq 20$–$23\, \mathrm{km\, s}^{-1}$

near the center (Baumgardt and Vasiliev, 2021; Nitschai et al., 2023). Where a single value is needed we adopt $\sigma = 21\,\mathrm{km\,s^{-1}}$, the midpoint of that central range and consistent with the oMEGACat 3D kinematic analysis (Häberle et al., 2025), and use it consistently in the influence radius of §2.3, the flyby velocities of §2.4, and Appendix A. The series' public calculators default instead to the cluster-averaged $18.2\,\mathrm{km\,s^{-1}}$ of the shared measurement compilation, which is a global rather than central-region figure; the difference propagates as a factor $(21/18.2)^2 \simeq 1.3$ in r_{infl} and is within the parameter variations of Appendix A.

The gravitational radius is

$$r_{\mathrm{g}} \equiv \frac{GM}{c^2} \simeq 3.0 \times 10^9\,\mathrm{cm} \left(\frac{M}{2\times10^4\,M_\odot}\right) \simeq 2.0 \times 10^{-4}\,\mathrm{AU} \left(\frac{M}{2\times10^4\,M_\odot}\right), \tag{1}$$

so the dynamic range between horizon scale and cluster scale is eight orders of magnitude in radius. The feasibility question is where in that range infrastructure can live.

2 The feasibility envelope

2.1 Constraint inventory

We consider a one-parameter family of architectures indexed by compactness: at one end, a dense swarm of independent elements on near-circular orbits at $r \sim 10^2$–$10^4\,r_{\mathrm{g}}$ (the configuration favored by the thermodynamic argument of Paper A, since proximity to the horizon minimizes entropy-transport losses); at the other, an extended bubble or shell of elements at $r \sim 10$–10^3 AU supported partly by radiation pressure or tether stress, the black-hole analogue of the Dyson bubbles whose collective stability was recently established by McInnes (2026). Intermediate cases (rings, nested tori) inherit constraints from both ends (Raval and Srikanth, 2024). We deliberately do not commit to an architecture; the envelope is the set of $(r, \text{architecture})$ pairs surviving all constraints simultaneously.

Six constraints bound the envelope: (i) relativistic orbit stability (inner); (ii) tidal stress on extended elements (inner); (iii) accretion-flow radiation and magnetic environment, if the hole is fueled (inner); (iv) cluster tidal truncation of bound orbits (outer); (v) stellar flyby perturbation and collision hazard (outer); (vi) thermal rejection capacity, which couples to the residue analysis of §3.

2.2 Inner boundary

Orbit stability. Circular equatorial orbits around a Kerr hole are stable outside the innermost stable circular orbit, $r_{\mathrm{ISCO}} = 6\,r_{\mathrm{g}}$ (Schwarzschild) shrinking to r_{g} (prograde, extremal) (Bardeen et al., 1972). For $M = 2\times10^4\,M_\odot$ this is $\sim 10^{-3}$ AU: dynamically, matter can orbit extraordinarily deep. Orbital periods are $P \simeq 2\pi(r^3/GM)^{1/2} \simeq 9$ s at the Schwarzschild ISCO and minutes at $100\,r_{\mathrm{g}}$ (619 s), and light-crossing latency across the swarm is seconds at the inner edge of the fueled envelope ($r/c = 9.8$ s at $100\,r_{\mathrm{g}}$), which is part of the computational attraction (Paper A, §4).

Tidal stress. A rigid element of size ℓ at radius r experiences differential acceleration $\sim 2GM\ell/r^3$. Requiring internal stress below a material strength S for an element of density ρ_{el} gives

$$\ell \lesssim \left(\frac{S\,r^3}{2GM\rho_{\mathrm{el}}}\right)^{1/2} \simeq 2\times10^2\,\mathrm{km} \left(\frac{S}{10\,\mathrm{GPa}}\right)^{1/2} \left(\frac{\rho_{\mathrm{el}}}{10^3\,\mathrm{kg\,m^{-3}}}\right)^{-1/2} \left(\frac{r}{10^2\,r_{\mathrm{g}}}\right)^{3/2} \left(\frac{M}{2\times10^4\,M_\odot}\right)^{-1}, \tag{2}$$

where 10 GPa is the tensile strength of present-day carbon composites. Tides therefore never forbid the swarm architecture (elements of km scale are unconstrained down to $\sim 10\,r_{\mathrm{g}}$); they forbid monolithic

structures below $\sim 10^3\, r_{\rm g}$ and thereby select swarms at small radii, a conclusion that parallels the single-star case (Wright, 2020).

Radiation and magnetic environment. If the hole is fueled at the level required for the Blandford–Znajek power budget of Paper A, the inner $\sim 10^2\, r_{\rm g}$ contains a magnetically arrested flow with field strength $B \sim 10^2$–10^4 G near the horizon (Tchekhovskoy et al., 2011; Narayan et al., 2022) and a jet along the spin axis. Equatorial swarm orbits outside the disk body ($r \gtrsim 10^2\, r_{\rm g}$, inclined to avoid both disk and jet) survive; structures inside $\sim 30\, r_{\rm g}$ face erosion by the flow itself. We adopt $r_{\rm in} \simeq 10^2\, r_{\rm g}$ as the practical inner boundary of a fueled configuration, and $r_{\rm ISCO}$ for a dormant one.

Formation coherence under relativistic precession. A constraint with no single-star analogue: swarm elements at different radii and inclinations precess differentially. Pericenter advance and Lense–Thirring nodal precession scale as $(r_{\rm g}/r)$ and $a_\star (r_{\rm g}/r)^{3/2}$ per orbit. At $r = 10^2\, r_{\rm g}$ the orbital period is $2\pi(r^3/GM)^{1/2} \simeq 619$ s $\simeq 10$ min, and a formation with a 2 per cent radial spread accumulates a differential *precession* phase of 2π in 1.7×10^3 orbits, about 12 days. Precession is not what binds, however: the same 2 per cent spread decoheres by ordinary Keplerian shear ($n \propto r^{-3/2}$) through 2π in 33 orbits, about 6 hours, a factor 50 faster at the same threshold. Keplerian shear also carries no radial dependence in units of the orbit, so it applies across the whole envelope rather than only inside $10^3\, r_{\rm g}$. The relativistic terms are what remain once a formation is built co-radially, which is the configuration that defeats shear by construction, and it is for that configuration that the numbered result below is stated. Because this exclusion recurs throughout the residue analysis, we state it as the paper's first numbered result:

> **Result 1 (decoherence).** Keplerian shear destroys any rigid shaped formation with a radial spread on timescales of hours, everywhere in the envelope; for co-radial formations, which are immune to shear, differential apsidal and Lense–Thirring precession does the same on day-to-week timescales inside $r \lesssim 10^3\, r_{\rm g}$. The deep envelope admits only axisymmetric shells, which are precession-invariant by symmetry, or phase-agnostic swarms that route function through communication rather than geometry. (3)

Rigid geometric formations (the flat rings and discs whose stability McInnes (2026) and Raval and Srikanth (2024) analyze around stars) are therefore excluded deep in the envelope. This is the Kerr-specific amendment to the single-star stability literature: the deep envelope selects swarm over monolith (tides, above), and within swarms, symmetric or phase-agnostic designs over shaped formations.

2.3 Outer boundary

Cluster truncation. The hole dominates the cluster potential inside its influence radius

$$r_{\rm infl} = \frac{GM}{\sigma^2} \simeq 0.2\,{\rm pc} \left(\frac{M}{2\times 10^4\, M_\odot}\right) \left(\frac{\sigma}{21\,{\rm km\,s^{-1}}}\right)^{-2} \simeq 4 \times 10^4\,{\rm AU}. \tag{4}$$

Orbits bound to the hole are progressively stripped by the fluctuating cluster field as $r \to r_{\rm infl}$; by analogy with tidal truncation of binaries in clusters (Binney and Tremaine, 2008; Heggie and Hut, 2003), we adopt $r \lesssim 0.1\, r_{\rm infl} \sim 4 \times 10^3$ AU as a stipulated prior edge for secure orbits, not a derived one. Binary truncation scales with the Jacobi radius, which for $2 \times 10^4\, M_\odot$ in the adopted uniform core ($\rho_0 = 3 \times 10^3\, M_\odot\,{\rm pc}^{-3}$) is $r_J = (M/4\pi\rho_0)^{1/3} = 0.81$ pc, or $4.15\, r_{\rm infl}$; the stipulated edge is then $0.024\, r_J$, and $0.1\, r_J$ would be 1.7×10^4 AU. The edge is a lever (Appendix C tabulates the downward direction), and the operative outer constraint below is stellar traffic rather than truncation.

Stellar flybys. The operative outer constraint is sharper than truncation: stellar traffic. The rate at

which cluster stars pass within impact parameter b of the hole, including gravitational focusing, is

$$\Gamma(b) = n_\star \, \pi b^2 \, v_{\rm rel} \left(1 + \frac{2GM}{b \, v_{\rm rel}^2} \right), \tag{5}$$

with $n_\star \simeq 10^4 \, {\rm pc}^{-3}$ (for mean stellar mass $0.3 \, M_\odot$ at the quoted mass density) and $v_{\rm rel} \simeq \sqrt{2}\,\sigma \simeq 30 \, {\rm km \, s}^{-1}$, the velocity used in the focusing term throughout. At $b = 10^3$ AU the focusing term is ~ 40 (it would be ~ 80 if evaluated at $v = \sigma$; we use $v_{\rm rel}$) and Eq. 5 gives one passage per $\sim 10^3$ yr; at $b = 10^2$ AU, one per $\sim 10^4$ yr; at $b = 10$ AU, one per $\sim 10^5$ yr (focusing-dominated regime, $\Gamma \propto b$). Each passage tidally perturbs swarm orbits at the impulsive level $\Delta v / v_{\rm orb} \sim 2(m_\star/M)\,(a/b)^2\,(v_{\rm orb}/v_{\rm rel})$ for structure semi-major axis $a < b$, saturating at $\sim 2(m_\star/M)(v_{\rm orb}/v_{\rm rel})$ for penetrating passages, where $v_{\rm orb} = (GM/a)^{1/2}$. The mass ratio controls everything: at $m_\star/M \sim 10^{-5}$ even a penetrating passage delivers $\Delta v / v_{\rm orb} \lesssim 10^{-2}$, so no single flyby disrupts a hole-bound orbit, and the hazard is cumulative, a random walk in eccentricity. Direct star–element scattering requires approach within $\sim 2Gm_\star/(v_{\rm rel} v_{\rm orb})$, a fraction $\lesssim 10^{-7}$ of the orbital cross-section per penetrating passage: negligible. And deep in the envelope the encounter duration $b/v_{\rm rel}$ exceeds the orbital period by orders of magnitude, so the impulsive estimate above is an overestimate: adiabatic invariance suppresses the coupling of slow perturbations to fast orbits exponentially (Binney and Tremaine, 2008; Heggie and Hut, 2003).

Secular cluster-tide forcing. Flybys are the granular part of the cluster field; the smooth part forces eccentricity secularly, the cluster analogue of Lidov–Kozai oscillations analyzed for binaries in cluster tides by Hamilton and Rafikov (2019). The secular timescale for a hole-bound orbit of semi-major axis a is of order $t_{\rm sec} \sim P_{\rm orb} \, M/M_{\rm cl}(a)$, where $M_{\rm cl}(a) = (4\pi/3)\rho_0 a^3$ is the smooth cluster mass enclosed by the orbit. Because the hole outweighs the enclosed cluster mass by $\gtrsim 10^5$ everywhere in the envelope, the forcing is weak: at $a = 10^3$ AU, $M_{\rm cl} \sim 10^{-3} \, M_\odot$ against $2 \times 10^4 \, M_\odot$ and $t_{\rm sec} \sim 3 \times 10^9$ yr, falling as $a^{-3/2}$ to $\sim 4 \times 10^8$ yr at the 4×10^3 AU stripping radius. The estimate assumes the tide's full quadrupole is available; ω Cen's measured near-spherical shape (global ellipticity ~ 0.1 from star-count isophotes, rounder still in the core; Geyer et al. 1983; Pancino et al. 2003; the photographic catalogue value is 0.17, White and Shawl 1987) suppresses the non-axisymmetric component that drives the largest eccentricity excursions, so these are upper limits on the forcing rate. Secular tides therefore do not bind interior to the stripping radius: they become comparable to the flyby-diffusion floor of §2.4 only at the envelope's outermost edge, where stripping already terminates it.

2.4 Monte Carlo of flyby histories

We quantify the cumulative hazard with a Monte Carlo over encounter statistics (Figure 1; method and parameters in Appendix A, code in the paper repository): impact parameters from the focused distribution of Eq. 5 within $b < 30a$, relative speeds drawn as described in Appendix A, impulsive kicks with the saturated form above, and dynamical survival defined as eccentricity random-walking to orbit-crossing ($e \sim 0.5$). The perturber mass function has two stellar components (main sequence plus white-dwarf tail; Häberle et al. 2024a) and, because kick variance scales as $m_\star^2$ and mass segregation concentrates heavy remnants inside the influence radius, a segregated remnant extension: neutron stars ($1.4 \, M_\odot$, 2 per cent number fraction) and stellar-mass black holes at a fiducial local number fraction of 1 per cent, bracketed by 0.1 and 3 per cent. We take the black-hole mass to be $31 \, M_\odot$, the mean mass of holes inspiraling into the IMBH in the González Prieto et al. (2025) ω Cen models, which identify $10 \, M_\odot$ as the assumption they are correcting; the metal-poor progenitors of a cluster at [Fe/H] ~ -1.5 leave heavier remnants than the solar-metallicity default. Caveats attach to this unbound-field comparison bracket, which retains the stipulated 0.1–3 per cent fraction: the inspiraling set is segregation-biased toward the heavy end, so the

population mean is lower than 31 $M_\odot$; and a bottom-heavy initial mass function of the kind those models require ($\alpha_3 = 2.5$) produces fewer black holes. The population that actually operates, the bound cusp channel below, is not assembled from separate figures; it is anchored jointly on the extended dark mass and the mean remnant mass directly (§2.4, Appendix A). At 31 $M_\odot$ per object, the 2–3 $\times 10^5$ $M_\odot$ extended dark component favored by the pulsar timing (Bañares-Hernández et al., 2025) is $\sim 10^4$ black holes, a global number fraction of 0.1–0.15 per cent against 7×10^6 cluster stars, consistent with the same bottom-heavy IMF and reaching a central, segregation-enhanced value of 1.15 per cent, close to the unbound bracket's own fiducial. The dataset that drives the $H_{\rm q}$–$H_{\rm sub}$ contest of §5 therefore also pins the Monte Carlo's dominant parameter.

The result is structurally simpler than the constraint inventory led us to expect: flybys never set the outer boundary, but the remnant tail controls the lifetime normalization. The median impulsive-diffusion lifetime is roughly flat from the ISCO to 4×10^3 AU, because the kick variance per encounter and the encounter rate scale inversely ($\langle \delta^2 \rangle \propto a^{-1}$ from the saturated penetrating passages that dominate it, $\Gamma \propto a$ in the focused regime), so their product is scale-free. Its value is $\sim 4 \times 10^9$ yr for the stars-plus-WDs function alone, dropping to $\sim 1 \times 10^9$, $\sim 2 \times 10^8$, and $\sim 9 \times 10^7$ yr at black-hole fractions of 0.1, 1, and 3 per cent. The perturber mass matters more than the fraction bracket: at fixed $f_{\rm BH} = 1$ per cent, moving from 10 to 31 $M_\odot$ raises $\langle m_\star^2 \rangle$ by a factor 8.0 and shortens the floor from $\sim 9 \times 10^8$ to $\sim 2 \times 10^8$ yr, a larger displacement than the whole 0.1–3 per cent bracket produces at fixed mass.

Adiabatic decoupling of the unbound channel. That floor is an impulsive estimate, and the impulsive approximation does not hold anywhere in the envelope. For the penetrating encounters that carry the kick variance, the adiabaticity parameter is $x = \omega_{\rm orb}\tau_{\rm enc} = (b/a)(v_{\rm orb}/v)$, which at $b \simeq a$ reduces to $v_{\rm orb}/v_{\rm rel}$: 2.2 at the cluster stripping radius, 4.5 at 10^3 AU, 14 at 10^2 AU, and 4×10^3 at the ISCO. Impulsive behaviour requires $x \ll 1$, which for this system means $a \gtrsim GM/v_{\rm rel}^2 \simeq 2 \times 10^4$ AU, five times outside the envelope. Unbound cluster stars are therefore adiabatically decoupled from every orbit an installation could occupy, and suppressing that correction is conservative in an unbounded way rather than by a stated factor. Applying the Gnedin and Ostriker (1999) kernel $A(x) = (1 + x^2)^{-5/2}$ per encounter, which is calibrated against N-body results and is the less suppressing of the standard choices, lifts the fiducial median to the 12-Gyr cluster-age cap within ~ 3 AU and to $\simeq 7.6 \times 10^8$ yr at the envelope edge, where the suppression is weakest. We carry both curves in Figure 1 and quote the impulsive one as the floor. The scale-free flatness is a property of the impulsive kernel, not of the physical problem: the corrected curve rises inward.

The Bahcall–Wolf cusp. The Monte Carlo uses the core-average perturber density, and a relaxed bound cusp around the hole raises it locally. The cusp exists. The González Prieto et al. (2025) realizations, which we cite elsewhere for growth history, report a density cusp inside the influence radius forming after 300 Myr, with the stellar population settling at $\gamma \simeq 1.3$ and the black-hole population steepening to $\gamma \simeq 2.0$: the multi-mass structure predicted by Bahcall and Wolf (1976, 1977), heavy species steeper than light. (The single-mass benchmark $\gamma \simeq 1.75$, reproducing the classic Bahcall–Wolf result inside the same ω Cen comparison, is Baumgardt et al. (2004)'s N-body value rather than a separate output of the multi-mass ω Cen realizations; we cite it for context but do not use it as an input.) The timescale argument agrees. ω Cen's catalogued relaxation times are ~ 4.0 Gyr in the core and ~ 12 Gyr at the half-mass radius (Harris, 1996); the influence radius lies deep inside the core radius, so the core value governs, and it is roughly a third of the cluster age. (A local Spitzer relaxation time evaluated at ρ_0 and σ with $\langle m \rangle = 0.3\,M_\odot$ gives ~ 13 Gyr instead, the convention difference residing in $\langle m \rangle$ and $\ln \Lambda$; we use the catalogued core value and flag the discrepancy.) Either way the segregation clock settles it: cusp formation for a species of mass m proceeds on $\sim (\langle m \rangle / m)\, t_{\rm relax}$, which for a 31 $M_\odot$ remnant is 0.03–0.6 Gyr on either convention.

The cusp case is therefore the fiducial, and it interacts with the adiabatic correction in the direction that matters. Cusp members are *bound*, moving at the local Keplerian speed rather than at $v_{\rm rel}$, so $x \sim 1$ for them and they are not adiabatically protected. That is the regime where the impulsive treatment is least controlled, not most: the impulsive limit wants $x \ll 1$ and the adiabatic limit wants $x \gg 1$, and the bound members sit between them. We apply the same Gnedin–Ostriker kernel there because it interpolates and is calibrated against N-body, but the correct statement is that the bound channel is the one where the kernel choice matters and the unbound channel is the one where the impulsive treatment is simply wrong. The price is measurable: running the bound channel with no kernel at all drops the conditional edge floor from 4.3×10^7 to 5.9×10^6 yr, a factor 7.4, and we carry that un-kerneled value as the lower bracket on this channel rather than quoting the kerneled number alone. Resonant relaxation is a further omission: for a near-Keplerian bound population it can dominate two-body relaxation in eccentricity, it is not in this model at all, and its absence pushes in the direction of longer lifetimes. Because $N(< r) \propto r^{3-\gamma} = r$ at $\gamma = 2$, only ~ 0.4 black holes are expected inside the stripping radius and ~ 4 inside $r_{\rm infl}$: the population is granular, not a smooth density, and usually absent at the edge, and we rebuild the Monte Carlo around it rather than estimate it analytically. Each species' cusp density follows $N_s(< r) \propto r^{3-\gamma_s}$ ($\gamma_\star = 1.3$, $\gamma_{\rm NS} = 1.5$, $\gamma_{\rm BH} = 2.0$; González Prieto et al. 2025), normalized against the core-average density at $r_{\rm infl}$; the perturber count for each history is Poisson-drawn from this profile rather than treated as a rate. At the envelope edge a bound black hole is present in 25 per cent of histories; when one is present the diffusion floor is $\sim 4 \times 10^7$ yr, and when none is present the channel reaches the 1.2×10^{10}-yr cluster-age cap, which is what the median history does. Three normalizations control those two numbers and each is stated rather than implied: the stipulated segregation enhancement of Appendix A (across 3–30 the presence probability spans 20–90 per cent), the joint dark-mass anchoring (Appendix A), and the choice to normalize the core number density on the observed *mass* density rather than on a fixed number density, which is worth a factor 2.7 in the perturber count on its own. Under the retired fixed-number convention (Appendix G), presence at the edge is 53 per cent and the median history is the one that grinds. The conditional floor itself barely moves between the two (4.3×10^7 against 4.3×10^7 yr): the normalization sets how often a bound remnant is there, not how fast it works when it is (Figure 1, panel b; Appendix A). That bimodal bound-member diffusion channel, not the earlier analytic estimate, is the binding clock at the envelope edge: flybys never bind anywhere in the envelope, and the deep envelope is limited by maintenance rather than by dynamics. The same remnant, held at $r_p = 4 \times 10^3$ AU, also forces the orbit secularly on $t_{\rm sec}(a) = (M/m_{\rm pert})(r_p/a)^3 P_{\rm orb}(a) \simeq 1.2 \times 10^7$ yr at $a \simeq 9 \times 10^2$ AU, but relativistic apsidal precession quenches the resonance ($t_{\rm sec}/t_{\rm GR} \propto a^{-4}$) inward of $a \simeq 4 \times 10^2$ AU, where $t_{\rm sec} \simeq t_{\rm GR} \simeq 3.7 \times 10^7$ yr; a competing quench from apsidal precession forced by the light cusp population is comparable near 10^3 AU and evaluated per history. Folding in the 77.5 per cent mutual-inclination window for the resonance, this forcing is coherent and oscillatory, not diffusive, so it sets a station-keeping cadence over its 4×10^2–4×10^3 AU operating range rather than a survival limit, and is absent below it.

The survival criterion itself needs a second clock. The $e = 0.5$ threshold marks dynamical death, when swarm orbits cross and membership is lost; an unmaintained swarm dies operationally much earlier, because differential kicks pump internal velocity dispersion, and once neighboring orbits cross at their much smaller spacing ($\Delta a/a \sim 10^{-3}$ for a dense swarm) the collisional cascade of §3.3 begins. Scaling the random walk to that threshold gives a grinding-onset time of $(e_{\rm swarm}/e_{\rm cross})^2 \simeq 4 \times 10^{-6}$ of the diffusion floor, $\approx 3 \times 10^2$ yr against the impulsive floor near 1 AU and longer once the adiabatic correction is applied: of order 10^2–10^3 yr, consistent with the abandonment timescale derived independently in §3.3. The two clocks are reconciled by maintenance: the station-keeping needed to null the accumulated differential drift is a small trim budget (the per-encounter kicks are $\Delta v \lesssim$ cm s^{-1} against orbital velocities of ~ 70 km s^{-1}

at the 4×10^3 AU stripping radius and $\sim 4 \times 10^3$ km s^{-1} at 1 AU), but it is strictly required. A passively bound swarm is not thereby functional; every long-lived swarm in this paper is a maintained swarm. The operative outer boundary remains the cluster stripping radius of §2.3, $\sim 0.1\, r_{\rm infl} \sim 4 \times 10^3$ AU, with the flyby Monte Carlo setting the maintenance-free lifetime interior to it.

2.5 The envelope

Combining §2.2–§2.4: for the fiducial system, engineered swarms (symmetric or phase-agnostic, per Result 1) persist passively from $r \simeq 10^2\, r_{\rm g}$ (2×10^{-2} AU, fueled) or $r_{\rm ISCO}$ (dormant) out to the cluster stripping radius $\sim 4 \times 10^3$ AU. The unbound field is adiabatically decoupled everywhere in that range; the operative floor is the bound-member diffusion channel, $\sim 4 \times 10^7$ yr at the envelope edge in the 25 per cent of realizations that draw a bound heavy remnant, and the cluster-age cap in the other three quarters. Beyond the stripping radius, orbits bound to the hole are not durable and only cluster-orbiting architectures (outside this paper's scope) remain.

The envelope is the paper's first result (Figure 1). Two corollaries matter downstream. First, the thermodynamically favored location (deep, near the flow) is also the dynamically safest, doubly so once adiabatic protection is counted: the environment does not penalize the architecture Paper A's physics prefers, which is a nontrivial consistency check the hypothesis could have failed. Second, the envelope concentrates any engineered mass within $\sim 4 \times 10^3$ AU $\sim 10^{-1}\, r_{\rm infl}$ of the hole, with the thermodynamic gradient pushing occupation deep into the arcsecond-unresolvable interior: observable only through its energetic and dynamical residue.

That is the subject of §3, after two remaining environmental questions: hazards other than flybys, and fuel.

2.6 Hazards beyond flybys

Three environmental processes remain to be treated quantitatively. Each is a candidate defeater; none defeats, though the first comes closest.

Tidal disruption events. A cluster-core IMBH tidally disrupts stars scattered into its loss cone. Rates for IMBHs in evolved globular clusters, computed with loss-cone methodology of the kind developed for the supermassive case (Stone and Metzger, 2016), are $\sim 5 \times 10^{-8}$–5×10^{-7} yr^{-1} per cluster for main-sequence stars (our evaluation of Fragione et al. (2018)'s Eq. 27 empty-loss-cone rate over the half-mass density bracket of its Eq. 8, 5×10^{-6}–5×10^{-5} yr^{-1} per IMBH-hosting cluster, reduced by an assumed IMBH occupation fraction of 0.01; the WD-suppression factor below is Fragione et al. (2018)'s stated result), with white-dwarf disruptions rarer by a factor of ~ 30 at $z \lesssim 0.5$, the epoch relevant here (the factor approaches ~ 100 only at $z = 2$) (Fragione et al., 2018); the ω Cen-specific value from the González Prieto et al. (2025) models is $\sim 5 \times 10^{-8}$ yr^{-1}, and we use that number where a target-specific rate is needed. One reciprocal flag for Paper A: this is also the natural loss-cone supply rate, and the staged architecture of Paper A's Appendix A requires $\sim$1–2×10^{-3} stars yr^{-1} during spin-up (6.8×10^4 stars over 4.2–6.6×10^7 yr, Paper A's own basis for the spin-up duration, which is a different quantity from the 3.5×10^7 yr Salpeter-time estimate of this paper's §5), so natural refilling falls short by four to five orders of magnitude and essentially the whole fuel budget of that phase must be delivered by engineered orbit-shaping. During a disruption the accretion luminosity approaches Eddington, $L \sim 3 \times 10^{42}$ erg s^{-1} for the fiducial mass, sustained for months to years of fallback: the flux at $r = 1$ AU is $\sim 10^9$ times the solar constant, and no plausible material hardening survives it in place. Feasibility therefore requires that a TDE be survivable by response rather than endurance: loss-cone stars are in principle identifiable and trackable long before disruption (the swarm sits at the bottom of the potential it would need to monitor), and the

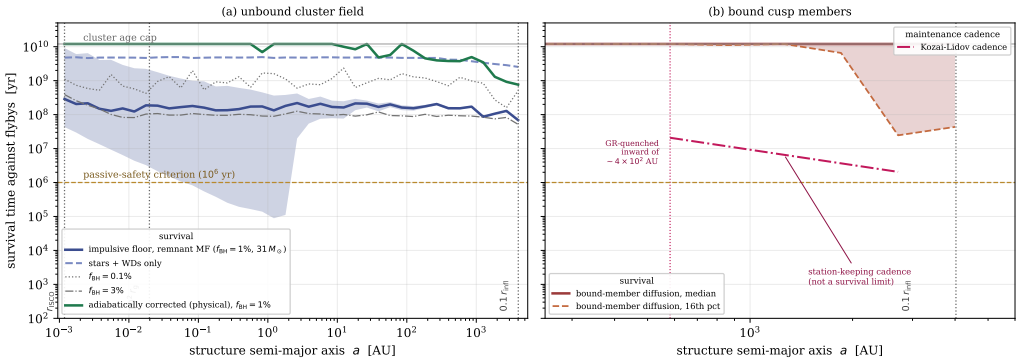


Figure 1: Survival time of hole-bound swarm structures against cumulative perturbation, from the rebuilt Monte Carlo of §2.4 (2×10^4 histories per radius bin; grid begins at the Schwarzschild ISCO). **(a) Unbound cluster field.** Blue curve and band: median and 16–84 per cent range of the impulsive estimate for the fiducial mass function with segregated remnants (31 $M_\odot$ black holes at 1 per cent number fraction, 1.4 $M_\odot$ neutron stars at 2 per cent); dashed: the stars-plus-WDs baseline; thin curves: black-hole fractions of 0.1 and 3 per cent. Green curve: the same fiducial case with the Gnedin and Ostriker (1999) adiabatic correction applied per encounter, which shows this channel is decoupled from every orbit in the envelope; it does not operate and is retained as the measure of that decoupling. **(b) Bound cusp members.** Dark red curve and band: median and 16–84 per cent range of the bound-member diffusion channel, jointly anchored perturber population ($f_{\mathrm{BH,central}} = 1.15$ per cent, 31 $M_\odot$); the population is Poisson-granular, so the 16th-percentile curve (dashed) is emphasized alongside the median because the distribution is bimodal by realization, not because the band is asymmetric noise. Magenta dash-dot: the Kozai–Lidov station-keeping cadence forced by the same bound remnant, drawn only over its GR-quenched-to-mass-quenched operating range and labeled on the curve as a maintenance requirement, not a third survival curve. This bound channel, not the decoupled field in panel (a), is the operative floor: flybys never bind anywhere in the envelope, and the deep envelope is limited by maintenance rather than by dynamics. Both panels: the band is a Monte Carlo precision interval, not a physical dispersion, and is narrow wherever the first-passage step count is large; vertical markers show the inner boundaries and the cluster stripping radius.

months-long fallback rise gives further warning, so temporary evacuation outward along the envelope, or shadowing at large inclination, is an engineering requirement we impose on any architecture rather than a reason to exclude one. The expected recurrence time of $\gtrsim 10^7$ yr then sets the natural amortization horizon of the installation: a 10^8–10^9-yr occupation must plan for several such events. Two observational corollaries. The TDE duty cycle also reconciles $H_{\rm eng}$ with the present silence at no extra cost: post-TDE fallback outshines any steady configuration for $\sim 10^2$–10^3 yr per $\sim 10^7$-yr recurrence, a duty cycle of $\lesssim 10^{-4}$, so catching ω Cen mid-flare was never likely under any hypothesis. And TDEs cut the other way, as an observational gift: any future TDE flare in ω Cen both confirms the hole and activates the variability channel of §3.2 at high signal-to-noise, and the framework of §4 treats a TDE with anomalous light-curve regulation as one of the few single-epoch events that can move $\ln K$ substantially.

Black-hole wander. The hole is a Brownian particle in the stellar bath: N-body characterizations give r.m.s. displacements of order 10^{-2}–10^{-1} pc for 10^3–$10^4\,M_\odot$ holes in ω Cen-like cores, extrapolating de Vita et al. (2018)'s fitted wander-amplitude scaling law (directly simulated over 75–150 $M_\odot$) to this mass range using its stated inverse-mass-power dependence. For the fiducial $2\times10^4\,M_\odot$ hole the expected wander is $\lesssim 10^{-2}$ pc $\sim 2\times10^3$ AU: comparable to the 4×10^3 AU envelope. That figure is a lower bound on two counts. The equilibrium amplitude in a multi-mass bath is set by the mass-weighted mean $m_{\rm eff} = \langle m^2\rangle/\langle m\rangle$ rather than by $\langle m\rangle$, since the velocity diffusion scales with $n\langle m^2\rangle$ while the drag scales with $n\langle m\rangle$ (Chatterjee et al., 2002; Merritt et al., 2007); for the segregated mass function of §2.4 that is $m_{\rm eff} = 13.1\,M_\odot$ against $\langle m\rangle = 0.75\,M_\odot$ at the fiducial $f_{\rm BH} = 1$ per cent and 31 $M_\odot$ perturbers, so an equal-mass estimate understates the amplitude by $\sqrt{m_{\rm eff}/\langle m\rangle} = 4.2$. Measured instead against a remnant-free bath ($m_{\rm eff} = 0.46\,M_\odot$), which is closer to the mass function the N-body results carried, the understatement is 5.4. Earlier drafts printed 2.3 against 0.54 (Appendix G); that pair is the 10 $M_\odot$ perturber run this paper retired at the referee-5 revision, and it understates the velocity diffusion, which scales as $m_{\rm eff}$ rather than as its root, by 5.8. And the quoted N-body results predate the retained black-hole populations now inferred for ω Cen, so the mass function they carried is lighter than the one §2.4 adopts. Paper C's wander program (§5.2 there) inherits both corrections, in the direction that makes the measurement easier. Structures bound to the hole simply ride along (the swarm orbits the hole, and hole plus swarm wander together through the cluster), so wander does not threaten the installation; what it threatens is the observer's astrometry, since the kinematic center and the hole need not coincide at the 10^{-2}-pc level. Paper C's astrometric programs already marginalize over center position; we note here that the wander amplitude is itself mass-dependent and therefore carries independent information about M in long-baseline proper-motion data.

Gas drag and erosion. ω Cen retains a measurable intracluster medium: pulsar dispersion measures in comparable clusters give central ionized densities $n_e \sim 0.07$–$0.3\,{\rm cm^{-3}}$ (Freire et al., 2001; Abbate et al., 2018). That range is the 47 Tucanae analogy of §2.7; Wang et al. (2025)'s own ω Cen figure is $1.94\,{\rm cm^{-3}}$, is flagged there as a foreground-dominated upper limit, and is not used, here or anywhere in this paper. Ram-pressure drag on a swarm element of areal density $\Sigma \sim 1\,{\rm g\,cm^{-2}}$ at orbital velocity of a few $\times10^3\,{\rm km\,s^{-1}}$ (the deep envelope; $4\times10^3\,{\rm km\,s^{-1}}$ at the 1 AU fiducial) removes a fractional momentum $\sim \rho_{\rm gas} v t/\Sigma$ per time t: at the measured densities this is $\lesssim 10^{-5}$ per kyr, running 1.9, 6.1, and 8.0×10^{-6} per kyr at $n_e = 0.07$, the fiducial 0.23, and $0.3\,{\rm cm^{-3}}$, and scaling as $n_e v/\Sigma$. The conclusion that drag never disrupts an orbit survives, but the term is not negligible over the envelope's own lifetimes: unopposed, the fiducial figure removes the full orbital momentum of a $1\,{\rm g\,cm^{-2}}$ element in 1.6×10^8 yr, which is the same order as the bound-member diffusion floor of §2.4. Drag therefore belongs in the maintenance budget rather than in the list of defeated processes, and an abandoned installation loses to it on a timescale the relic window of §3.3 already assumes. Sputtering and dust impacts are similarly small in an old cluster with no star formation and depleted debris populations. Gas matters for fuel, however, which is the next

subsection.

2.7 The fuel budget

The Blandford–Znajek architecture of Paper A needs mass supply, and the measured medium bounds it. ω Cen has no direct ionized-gas density measurement of its own; we adopt $n_e \simeq 0.23\,\mathrm{cm}^{-3}$ by analogy with the comparable cluster 47 Tucanae (Abbate et al., 2018), an explicit ambient-value choice rather than an ω Cen-specific measurement. (Wang et al. 2025's own ω Cen-specific figure, $n_e = 1.94\,\mathrm{cm}^{-3}$, is $\sim 8\times$ higher and is flagged by its authors as an upper limit likely dominated by foreground, non-intracluster ionized gas rather than the cluster's own medium; we do not use it here.) Bondi accretion from gas at this density and relative velocity $\sim \sigma$ gives

$$\dot{M}_{\mathrm{B}} = \frac{4\pi (GM)^2 \rho_{\mathrm{gas}}}{(\sigma^2 + c_s^2)^{3/2}} \simeq 3 \times 10^{18}\,\mathrm{g\,s^{-1}} \left(\frac{M}{2\times 10^4\,M_\odot}\right)^2 \left(\frac{n_e}{0.23\,\mathrm{cm^{-3}}}\right) \simeq 5\times 10^{-8}\,M_\odot\,\mathrm{yr}^{-1}, \qquad (6)$$

about 10^{-4} of the Eddington rate. The figure is bounded from both sides. It is an upper bound insofar as the density at the hole's actual location may be below the cluster mean: the accretion non-detections are consistent with the hole occupying a locally evacuated region (Mahida et al., 2026), and the Bondi radius samples gas the surveys average over; quoted from the same denominator as Eq. 6, $GM/(\sigma^2 + c_s^2) \simeq 3\times 10^4$ AU. It is a lower bound insofar as it ignores harvesting, though by far less than earlier versions of this paper and of Paper F claimed (Appendix G). Those versions put $\sim 10^3$ giants inside the influence radius; that figure counted the stars there rather than the ones currently on the giant branch. Recounting with the Kroupa turnoff weight (0.035) and the fraction of a turnoff star's main-sequence life spent above $10^3\,L_\odot$ (0.032), the expected standing giant population inside this paper's own $r_{\mathrm{infl}} = 0.20$ pc is 0.18, rising to 1.4 if the same radius is populated at Paper F's central density and to 8.4 inside the larger influence radius of Paper F's $4\times 10^4\,M_\odot$ hole. At $10^{-8}\,M_\odot\,\mathrm{yr}^{-1}$ each, the aggregate wind supply is 3 per cent of $\dot{M}_{\mathrm{B}}$ on this paper's own numbers and at most 1.6 times it on Paper F's. Local wind harvesting is therefore a factor-of-order-unity term, not an orders-of-magnitude one, and the "orders of magnitude" route is deliberate mass import alone, which the waste-heat ceiling of §3.1 already prices. At MAD-plus-spin effective efficiencies of order unity, realized only at high spin ($\eta_{\mathrm{jet}} \simeq 1.4$ at $a_\star \simeq 0.99$, falling roughly as $a_\star^2$ to a factor ~ 5 lower at $a_\star = 0.5$; Paper A; Tchekhovskoy et al. 2011), the ambient rate gives extractable power $P \sim \dot{M}_{\mathrm{B}} c^2 \simeq 3\times 10^{32}$ W $\simeq 7.6\times 10^5\,L_\odot$ at $a_\star \simeq 0.99$ and $\approx 9.5\times 10^2$ to $3.8\times 10^3\,L_\odot$ at the MAD equilibrium spin $a_\star = 0.035$–0.07 of §3.4, with the fuel ceiling carrying the same $\eta(a_\star)$ dependence: on the thin-disc branch the ambient medium alone, with no harvesting of stellar winds or imported fuel, funds a computational budget two orders of magnitude above the waste-heat ceiling derived in §3.1, and on the MAD branch it does not (§3.4).

That headroom assumes the Bondi rate reaches the horizon, and in natural hot, low-Eddington flows it does not: the inflow declines inward, $\dot{M}(R) \propto R^s$ with $s \simeq 0.3$, as most of the captured gas is unbound into outflows (Blandford and Begelman, 1999; Yuan and Narayan, 2014). Over the eight decades from the Bondi radius ($r_{\mathrm{B}} = GM/(\sigma^2 + c_s^2) \simeq 3.3\times 10^4$ AU, Eq. 6) to r_{g}, the suppression is $(r_{\mathrm{B}}/r_{\mathrm{g}})^{0.3} \approx 290$, giving $P_{\mathrm{fuel}} \simeq 2.6\times 10^3\,L_\odot$; adopting instead the empirical Sgr A* figure of ~ 1 per cent of the Bondi-radius rate gives $\simeq 8\times 10^3\,L_\odot$. Either value lands at or below the warm-swarm waste-heat ceiling of §3.1, so for a natural flow the supply and concealment constraints bind together, and concealment alone binds only in the engineered case. The engineered case is available, and it unifies two of this paper's results: a system drawing more than the ADIOS-suppressed rate is a system that has suppressed its own outflow, and that is the same control action, regulating horizon magnetic flux and mass supply, that produces the low-R variability deficit of §3.2. The fuel budget and the R statistic are two faces of one control

action. We therefore carry both cases through the paper: natural-suppressed ($P_{\rm fuel} \sim 10^3$–$10^4\,L_\odot$) and engineered-unsuppressed ($P_{\rm fuel} \simeq 7.6 \times 10^5\,L_\odot$), with the Appendix C prior ceiling stated for the pair.

For an installation with outflow control, then, the binding constraint on a present-day ω Cen system is thermal (getting rid of entropy without detection), never supply; without it, the two constraints meet. The unsuppressed ordering is a nontrivial output: for stellar-mass holes in the same environment the two constraints reverse even in the engineered case, which is an independent reason the hypothesis family selects IMBHs. The same numbers close the loop on the accretion non-detection: the deep ATCA and JWST limits require the natural radiative efficiency of any Bondi-fed flow to be $\lesssim 4 \times 10^{-3}$ (Mahida et al., 2026; Chen et al., 2026), which is uncomfortable for $H_{\rm q}$ at the upper end of the mass range and what $H_{\rm eng}$ predicts for a flow whose output is being extracted as work rather than radiation; the inverse fundamental-plane treatment of the same candidate reaches the complementary conclusion, that its radio silence constrains gas supply, accretion efficiency, and compact-core production rather than excluding the hole (Abbassi et al., 2026). The adjudication in §5 prices this observation for both hypotheses rather than letting either claim it informally.

3 Observable residue

3.1 Waste-heat floor

Paper A argued that horizon entropy disposal permits computation with radiated waste heat far below the Dyson-sphere expectation. Here we make the floor quantitative, and we first fix the bookkeeping, because the mainstream position is an energy-conservation argument: Curtis et al. (2026) state that Dyson-scale computation (the stellar-energy regime of the Kardashev scale; Kardashev 1964) must reradiate nearly all of the energy it absorbs, making mid-infrared thermal emission the robust technosignature. The horizon-sink architecture disputes that position at the level of energy conservation itself, not merely entropy accounting. Let $P_{\rm comp}$ be the power processed by the swarm and $f_{\rm sink}$ the fraction of the waste *energy* delivered across the horizon rather than radiated; the entropy rides with its carriers, so a single fraction serves for both. Energy beamed across the horizon is not reradiated at all: it adds to M, and since $T_H \propto M^{-1}$ the disposal channel deepens as it is used, a weakly self-improving property. What must be radiated is only the missed fraction, $L_{\rm waste} = (1 - f_{\rm sink})\,P_{\rm comp}$, re-emitted thermally at the temperature set by the swarm radius. For single-sided radiators at unit covering fraction, $T_{\rm eff} \simeq (L_{\rm waste}/4\pi r^2 \sigma_{\rm SB})^{1/4}$: $1\,L_\odot$ at $r = 1$ AU gives 394 K, $1\,L_\odot$ at 10^3 AU gives 12.5 K, and a 50 K swarm at 10^3 AU corresponds to $L_{\rm waste} \approx 260\,L_\odot$. This $T_{\rm eff}(r)$ assumes unit covering fraction with single-sided radiators; a sparse swarm of covering fraction $f_{\rm cov}$ runs hotter by $f_{\rm cov}^{-1/4}$, moving mass into the JWST wedge and shifting the composite $\ln K$ of §5 to -0.61 at $f_{\rm cov} = 10^{-2}$, with the excluded active-prior fraction rising from 0.76 to 0.92 (Appendix C, `--fcov`).

The engineering limit on $f_{\rm sink}$ is transport: waste energy must be carried inward, against the swarm's own power draw, by mass flux or directed radiation into the horizon. The capture-cone and étendue accounting of Appendix A.3 bounds $1 - f_{\rm sink} \gtrsim 10^{-4}$ for swarm architectures within the envelope, set by the fraction of beamed power unavoidably intercepted and re-thermalized by swarm elements along the transport path. The sign of its influence is worth stating: a lower true floor (better beaming than the A.3 accounting allows) weakens the mid-infrared charge against $H_{\rm eng}$ and moves $\ln K$ toward zero, while a higher floor tightens the $P_{\rm comp}$ ceiling and strengthens it (§6). The floor is therefore

$$L_{\rm waste} \gtrsim 10^{-4}\,P_{\rm comp}. \tag{7}$$

Converting that floor into a $P_{\rm comp}$ bound requires a temperature axis, because which instrument limits $L_{\rm waste}$ depends on where the thermal emission peaks. The deep JWST limits at the kinematic center (Chen

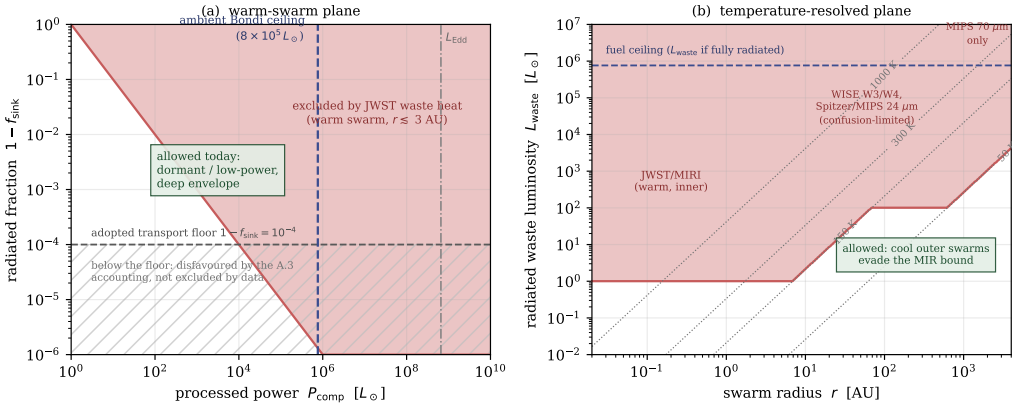


Figure 2: Constraint planes for an ω Cen installation. **(a)** The ($P_{\rm comp}$, $1 - f_{\rm sink}$) plane for a warm (inner-envelope) swarm. Red: excluded by the JWST mid-infrared point-source limit. Grey hatching below the dashed grey line: the adopted entropy-transport floor $1 - f_{\rm sink} = 10^{-4}$, a parameter carried over from the Appendix A.3 accounting rather than a data-driven boundary (§6). Blue dashed: the ambient Bondi fuel ceiling ($7.6 \times 10^5\, L_\odot$, the engineered-unsuppressed case; the ADIOS-suppressed natural supply is $\sim 2.6 \times 10^3\, L_\odot$, §2.7; imported mass moves it right at the cost of visibility). **(b)** The temperature-resolved (r, $L_{\rm waste}$) plane: dotted lines are $T_{\rm eff}$ contours, and the excluded wedge is the radius-dependent instrument limit (JWST/MIRI for warm inner swarms; WISE W3/W4 and Spitzer/MIPS, confusion-limited in the core, for cool outer ones). The surviving region is dormant, low-power, or cool and outer.

et al., 2026) constrain warm sources: a swarm at $r \lesssim$ a few AU re-radiates at $T_{\rm eff} \gtrsim 150$ K, peaks within MIRI coverage, and inherits the sub-$L_\odot$ point-source limit, giving $P_{\rm comp} \lesssim L_{\rm lim}/(1 - f_{\rm sink}) \lesssim 10^3$–$10^4\, L_\odot$. A cool outer swarm evades it: at $r \sim 10^2$–10^3 AU the emission peaks at 20–60 μm and beyond, past the coverage of JWST/MIRI's nine imaging filters (F560W through F2550W, the 25.5 μm cutoff); WISE W3/W4 and Spitzer/MIPS, whose 6–18 arcsec beams are confusion- and crowding-limited in the ω Cen core, are the instruments that would bound such a cool swarm observationally, though Appendix C does not draw on them directly. The defensible statement is radius- and temperature-dependent, and the continuous per-filter Appendix C likelihood (§C) integrates the two-temperature swarm SED against the real MIRI sensitivity curve rather than marginalizing over instrument-specific placeholder limits, so a sufficiently cool, extended swarm falls below every MIRI filter and registers as undetected rather than excluded. An engineered system operating at the radius-appropriate ceiling is consistent with all current data; deeper MIR photometry tightens the warm wedge linearly, while the cool wedge waits on far-infrared sensitivity. Figure 2 assembles the warm-swarm plane and the temperature-resolved plane, with the transport floor and the fuel ceiling of §2.7, into the constraint space that defines the surviving $H_{\rm eng}$ parameter volume, the single-system counterpart of the Ĝ survey's galaxy-scale waste-heat plane (Wright et al., 2014b,a).

3.2 The MAD-regulation signature

The second channel is new to this paper. General-relativistic magnetohydrodynamic simulations establish that natural accretion in the magnetically arrested state is intrinsically episodic: magnetic flux accumulates at the horizon, chokes the inflow, and erupts in quasi-regular flux-expulsion events, producing characteristic variability in jet power and radiative output with a MAD > intermediate > SANE hierarchy in both luminosity and its variance (Tchekhovskoy et al., 2011; Narayan et al., 2022; Raha et al., 2026). The eruption mechanism is plasmoid-mediated reconnection of the accumulated horizon flux, resolved ab initio by Ripperda et al. (2022), whose flare cycle of $\sim 10^2$–$10^3\, r_{\rm g}/c$ between flux accumulation and

expulsion is the recurrence we adopt; the state-resolved variability tables of Raha et al. (2026) place the MAD eruption power in the same interval at matched mean luminosity. For $M = 2 \times 10^4\, M_\odot$ that cycle is $\sim$ 10–10^2 s, comfortably within the cadence of X-ray monitoring.

A system engineered for steady power extraction has an incentive absent in nature: eruptions are interruptions. Regulating the delivered mass and flux (the control variable identified by the simulations is horizon magnetic flux itself) suppresses the eruption cycle and its variability signature. We therefore forward-model the residue as a *variability deficit*: an accreting IMBH whose X-ray (or optical) power spectrum lacks the flux-eruption band that GRMHD baselines predict for its luminosity and inferred state. Quantitatively, we define the regulation statistic R as the ratio of integrated power in the eruption band (10^{-2}–10^{-1} Hz for the fiducial mass) to the GRMHD-calibrated expectation at matched mean luminosity; natural MADs populate $R \sim 1$ with factor-of-few scatter, and $R \ll 10^{-1}$ sustained over many cycle times has no identified natural mechanism (Figure 3). The band edges follow directly from scaling the Ripperda et al. (2022) flare cycle of 10^2–$10^3\, r_g/c$ to $M = 2 \times 10^4\, M_\odot$ ($r_g/c = 0.1$ s, hence 10–100 s), with the Raha et al. (2026) state tables placing the eruption power in the same interval. The main confusion risk in applying the statistic is contamination from the intermediate magnetic state, whose eruption power is partially suppressed relative to MAD, so the baseline must be conditioned on the inferred state, never on luminosity alone.

Two instrument realities shape where R can be measured. First, at the fiducial mass R is an X-ray/optical statistic; radio is excluded twice over. Synchrotron self-absorption in any compact GHz-emitting flow puts the $\tau = 1$ photosphere at $\sim 10^2$–$10^4\, r_g$, low-pass filtering intrinsic variability at $\Delta t \gtrsim$ 10–10^3 s, directly on top of the eruption band, so a radio non-detection of the band is expected under H_q and H_{eng} alike and carries no discriminating power. Interstellar scintillation compounds the problem: a microarcsecond-scale source at 5.49 kpc scintillates at GHz frequencies on minutes-to-hours timescales, adding modulation uncorrelated with accretion in and near the band, so even a genuinely regulated (low-R) source would show a measured R biased upward unless the scintillation is modeled out. Second, the statistic has a detectability threshold. Distinguishing $R \sim 1$ from $R < 0.1$ requires several counts per $\sim$ 10 s cycle sustained over many cycles, i.e., $F_X \gtrsim$ a few $\times 10^{-12}$ erg cm^{-2} s^{-1} for m^2-class effective area, corresponding to $L_X \gtrsim 10^{34}$–10^{35} erg s^{-1} $\approx 4 \times 10^{-9}$–$4 \times 10^{-8}\, L_{Edd}$ at the fiducial mass. The channel therefore activates in flare and TDE states rather than in deep quiescence, which is consistent with the TDE-as-gift framing of §2.6: the events that confirm the hole are the events bright enough to measure R.

The signature has three properties valuable for adjudication. It is conditional: it activates only if accretion is ever detected, so it presently constrains nothing (ω Cen's hole is electromagnetically silent; Mahida et al. 2026). It is differential: it compares a source against a physics baseline at matched parameters, canceling many systematics. And it is disprovable in place: detection of normal flux-eruption variability from a future ω Cen accretion flare would count against H_{eng} in the framework below, making the channel one of the few that can move evidence in both directions.

3.3 Relic residue: the abandoned swarm

Persistence is conditional on maintenance, and Lacki (2025) has shown what maintenance failure means for megastructures: once guidance fails, swarm elements collide, and a collisional cascade grinds the population to dust on a timescale of roughly the orbital period divided by the covering fraction. The deep envelope makes this dramatic. At $r \sim 1$ AU around the fiducial hole the orbital period is $\sim$ 3 days; for covering fractions 10^{-4}–10^{-2} the first-collision timescale $\sim P/f_{cov}$ is years to decades, and the full cascade completes on a timescale of order 10^3 yr, orders of magnitude faster than for the stellar-orbit megaswarms Lacki considered and consistent with the grinding-onset clock of §2.4. The end state differs too, and the difference is observationally decisive: cascade debris around a star settles into a long-lived

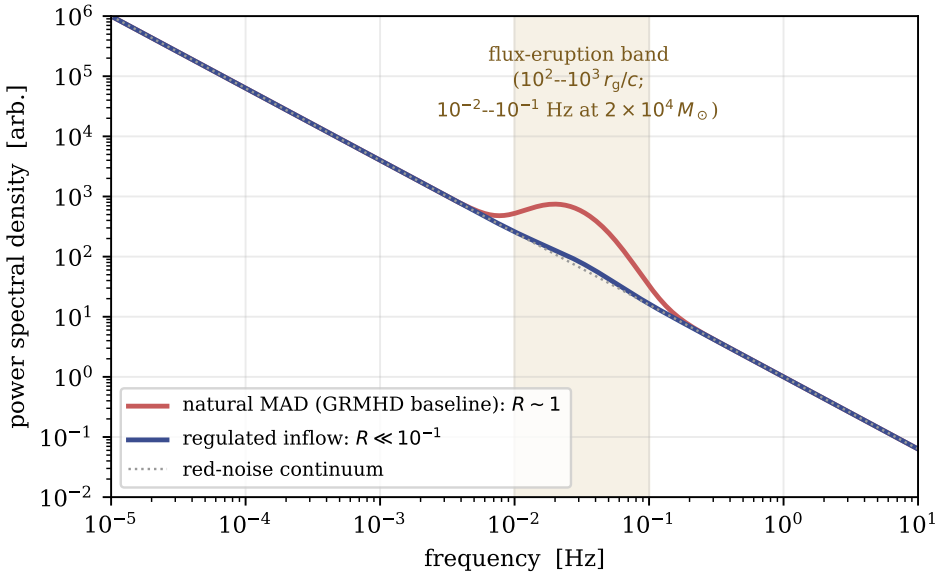


Figure 3: Schematic definition of the regulation statistic R: power spectra of an accreting IMBH at matched mean luminosity under natural MAD accretion (flux-eruption band populated, $R \sim 1$) and regulated inflow ($R \ll 10^{-1}$). Band location scales inversely with black-hole mass; values shown for $M = 2 \times 10^4\,M_\odot$.

warm dust population, a passive relic technosignature, whereas debris around a black hole is progressively drained, its periapsis distribution fed toward the loss cone by continuing collisions and flyby perturbations and its finest grindings coupled to whatever gas flow exists; the drainage rate remains an estimate, qualified in §6. The hole removes its own debris. An abandoned engineered-IMBH system therefore passes through a brief bright phase (cascade grinding plus enhanced accretion luminosity as debris drains) and then reverts to a naked quiescent hole, indistinguishable from $H_{\rm q}$ except possibly through spin.

Three consequences follow. First, a fourth hypothesis joins the menu at zero structural cost: $H_{\rm eng}^{\rm relic}$, a formerly engineered now-abandoned system, whose observables today equal $H_{\rm q}$ plus high spin, and whose prior couples to the goal-stability open problem flagged in Paper B. Second, the searchable relic window is short, so population-level searches for dead installations (the analogue of the dust-relic searches now proposed for stellar systems) have low yield around IMBHs specifically; non-detection of relic dust in ω Cen carries almost no evidence either way, and the framework scores it accordingly. Third, spin becomes the only durable observable, and an objection must be met before relying on it: Blandford–Znajek extraction spins the hole down, and MAD jets remove angular momentum faster than accretion supplies it (Narayan et al., 2022), so one might expect long use to erase the very fossil we propose reading. The rates rescue the argument: in the MAD spin-down calculus of Narayan et al. (2022), the spin-down per unit accreted mass is of order $|\Delta a_\star| \sim \Delta M/M$, so erasing a high spin requires cycling a mass comparable to the hole's own through the flow, and at the ambient fueling of §2.7 that takes $M/\dot{M}_{\rm B} \sim 4 \times 10^{11}$ yr. Only a civilization importing mass at far above ambient rates for $\gg 10^9$ yr would measurably despin the hole, and that regime is separately excluded by the waste-heat ceiling. This is also the point of contact with the compact-object signature named in the review literature: Vidal et al. (2026) attach to Penrose-process extraction from a Kerr hole the signature of "a black hole whose rotational rate is slowing down anomalously". The calculus above prices that signature: the anomalous case requires the extraction or mass-import regime, far above ambient fueling for $\gg 10^9$ yr, that the waste-heat ceiling of §3.1 excludes. Anomalous spin-down is diagnostic of that regime wherever it occurs, and at ω Cen the ceiling predicts its

absence for any installation still concealed.

The spin channel needs a careful statement, including an isotropy assumption. The natural expectation is a channel mixture, set by growth history rather than by any single attractor: repeated comparable-mass mergers drive χ toward the $\simeq 0.7$ attractor (Fishbach et al., 2017; Gerosa and Berti, 2017); gas-poor growth by minor mergers and inspirals of compact objects random-walks the spin to low values under isotropic capture; and prolonged coherent disk accretion spins the hole up toward the radiation-limited maximum $a_\star = 0.998$ (Bardeen, 1970; Thorne, 1974). Which of the three operated at ω Cen is a question its growth models answer, conditional on the isotropy assumption stated next.

For the minor-merger regime, isotropic capture ($\langle\cos\iota\rangle = 0$ for the captured objects' orbital planes) gives $a_{\star,\mathrm{rms}} = (\tilde{L}/\sqrt{3})\sqrt{\mu/M_f}$ (Appendix B); the assumption is load-bearing for everything that follows and we state it here rather than leave it implicit. A net alignment fraction $\epsilon_{\mathrm{rot}} = \langle\cos\iota\rangle$ above isotropic adds a coherent term $a_\star^{\mathrm{coh}} \approx (1.7\text{–}2.0)\,\epsilon_{\mathrm{rot}}$ that carries no $\sqrt{N}$ suppression, so it matches the random walk at $\epsilon_{\mathrm{rot}} \simeq 0.026\text{–}0.031$ (Appendix B) and dominates above it, erasing the low-spin prediction entirely once $\epsilon_{\mathrm{rot}} \gtrsim 0.4\text{–}0.5$. The prediction below holds only for $\epsilon_{\mathrm{rot}} \lesssim 0.03$. ω Cen is a rotating cluster and the inspiraling population is the mass-segregated heavy-remnant tail, so isotropy is a premise to check rather than a safe default; $\langle\cos\iota\rangle = 0$ is an assumption here, and the González Prieto et al. (2025) realizations cannot supply it, since their Monte Carlo code is spherically symmetric, does not track the orbital plane, and draws transverse directions isotropically. §3.3 carries the tolerance as an explicit condition on the green branch.

The González Prieto et al. (2025) realizations (cited here for growth history only; they do not track spin) grow a 500–5000 $M_\odot$ seed to $\sim 5\times10^4\,M_\odot$ primarily by mergers with 30–40 $M_\odot$ black holes, roughly 10^3 events at seed-epoch mass ratio $q \simeq 0.006\text{–}0.08$; by the end of the growth history the ratio the final spin depends on is the smaller $\mu/M_f \simeq 7\times10^{-4}$. That is the isotropic minor-merger regime, not the comparable-mass one: the walk reaches the $\chi \simeq 0.7$ attractor only at $q \approx 0.12$ (Appendix B, a continuous scaling rather than a literature-sourced cutoff), and ω Cen sits a factor ~ 175 below that boundary on the final-mass ratio. So the $\chi \simeq 0.7$ attractor of Fishbach et al. (2017) and Gerosa and Berti (2017) does not apply to this target; under isotropic capture the expected outcome, derived rather than imported from a comparable-mass literature value (Appendix B), is $a_\star \sim 0.05\text{–}0.10$: 0.049 ± 0.021 at the González Prieto et al. (2025) endpoint mass, 0.084 at this paper's fiducial mass. For ω Cen the contrast is the sharper low-versus-high one, and the attractor caution belongs to the class-level statement rather than to this system. The information forecast of §5 restores the spin channel accordingly, with the caveat that the class-level application to the LISA IMBH population must carry the mixture, since comparable-mass growth histories elsewhere populate the attractor.

3.4 Which accretion architecture spun the hole up

The fuel and power budget of §2.7 quotes $\eta_{\mathrm{jet}} \simeq 1.4$ at $a_\star \simeq 0.99$, which is a magnetically arrested flow at near-extremal spin. The spin-up arithmetic of §5 assumes Thorne's radiative equilibrium $a_\star = 0.998$, a thin-disc boundary condition. These are two accretion architectures, distinguished by how the hole was spun up. Extraction at the ambient fueling rate preserves a thin-disc spin: by the despin calculus of §3.3, erasing a high spin requires cycling a mass comparable to the hole's own through the flow, which takes $M/\dot{M}_{\mathrm{B}} \sim 4\times10^{11}$ yr at ambient fueling. A hole spun up on a thin disc therefore keeps its spin for far longer than a Hubble time while extracting at $\eta \sim 1$. The fork lies upstream, in the spin-up itself. On a thin disc the hole climbs to $a_\star = 0.998$. In a magnetically arrested flow the jet removes angular momentum faster than accretion supplies it above the arrested equilibrium, so spin-up from a low spin stalls at the arrested equilibrium; Narayan et al. (2022) (§3.4) find the spin-up and spin-down equilibrium at $a_\star^{\mathrm{eq}} \approx 0.035$, note that the shorter simulations of Tchekhovskoy et al. (2012) gave the larger ≈ 0.07,

and speculate that $a_\star^{\rm eq} \to 0$ for sufficiently long-lived MAD systems. We carry $a_\star^{\rm eq} = 0.035$–0.07 for the MAD-mode spin-up (earlier versions quoted 0.1–0.3 for this equilibrium; Appendix G). Every number downstream of the 2.20× growth factor inherits the thin-disc assumption.

We therefore state the fork and price both branches (`figs/fE_spin_branch.py`) rather than choosing one silently. On the *thin-disc branch*, the episode is a spin-up to $a_\star = 0.998$ with no sustained extraction during it, giving $M_f/M_i = 2.2024$, $\Delta M = 2.4 \times 10^4\, M_\odot$ from the fiducial hole, an episode lasting 3.6×10^7 yr, and endpoints of $4.4 \times 10^4\, M_\odot$ from the fiducial mass or $1.1 \times 10^5\, M_\odot$ from the merger endpoint. On the *MAD branch*, the terminal spin is the equilibrium value and the same Bardeen relation, $a_\star = \sqrt{2/3}\,(M_i/M)\left[4 - \sqrt{18(M_i/M)^2 - 2}\right]$ from $a_\star = 0$, gives $M_f/M_i = 1.0097$ at $a_\star = 0.035$ and 1.0198 at $a_\star = 0.07$: $\Delta M = 1.9 \times 10^2$ to $4.0 \times 10^2\, M_\odot$, episodes of 0.43 to 0.88 Myr, and endpoints of 5.05 to $5.10 \times 10^4\, M_\odot$ from the merger endpoint. These MAD-branch values follow from the closed relation; the script (`figs/plc_emix/spin_branch_aeq.py`), run at $a_\star = 0.035$ and 0.07, returns the same values to the digits quoted.

Three of this paper's observables depend on which branch holds. The mass discriminant of §5 exists only on the thin-disc branch; on the MAD branch the predicted endpoint is within 2 per cent of the merger-only mass and the channel carries no information at all. Core depletion likewise: $\Delta M \sim 10^2\, M_\odot$ is undetectable against any cluster mass model. And the durable residue $a_\star \gtrsim 0.9$, which §3.3 makes the one lasting observable of an engineered history, becomes $a_\star \approx 0.035$–0.07. That range lies within 1σ of the natural-capture null of Appendix B (0.049 ± 0.021 at the endpoint mass; 0.084 at the fiducial mass) and below the T5 trigger $a_\star < 0.1$. The spin contrast therefore vanishes or inverts: 0.035 lies below the central value of the null and 0.07 above it, both within 1σ, so a measurement would struggle to separate an engineered hole from a natural one, and at 0.035 the engineered hole reads lower. It follows that the spin test T5 of Paper A, which kills the exploitation model at $a_\star < 0.1$, would fire on an engineered hole that was spun up in MAD mode, so T5 is conditional on the thin-disc branch.

Nor is the fuel ceiling independent of the choice. §2.7's $7.6 \times 10^5\, L_\odot$ is the ceiling at $a_\star \simeq 0.99$ and carries the $\eta(a_\star) \propto a_\star^2$ dependence stated there, so it scales by $(a_\star/0.99)^2$. At $a_\star = 0.035$ that factor is 1.25×10^{-3}, giving a ceiling of $\approx 9.5 \times 10^2\, L_\odot$; at $a_\star = 0.07$ it is 5.0×10^{-3}, giving $\approx 3.8 \times 10^3\, L_\odot$. They bracket the ADIOS-suppressed natural supply ($\sim 2.6 \times 10^3\, L_\odot$), each within a factor of 3 of it; §2.7 places that supply at or below the warm-swarm waste-heat ceiling of §3.1. The engineered case's margin of two orders of magnitude over that ceiling is a thin-disc-branch statement and does not carry to the MAD branch.

We do not adjudicate the fork here, because the observation that would is a spin measurement, and it does not exist. What changes is the bookkeeping: every claim in §3 and §5 that descends from 2.20, from $1.1 \times 10^5\, M_\odot$, from 3.5×10^7 yr, or from $a_\star \gtrsim 0.9$ is a thin-disc-branch claim, and is labelled as one from here on. A referee who prefers the MAD architecture should read those numbers as the upper branch of a fork whose lower branch removes the mass channel and the depletion signature, inverts the spin contrast, and lowers the ambient fuel ceiling by a factor of ≈ 200 to 800.

3.5 EMRI dephasing: the mass channel

Every residue channel above constrains power, variability, or spin. One channel constrains engineered *mass*, and it comes for free with the gravitational-wave observation the series already relies on. Matter in the vicinity of an inspiral imprints secular phase shifts on the waveform; the environmental-effects literature for extreme- and intermediate-mass-ratio inspirals, founded by Barausse et al. (2014), quantifies how accretion disks, dark-matter spikes, and other mass distributions dephase an inspiral over an observation. An engineered swarm is a mass distribution like any other. If a stellar-mass compact object is ever caught

inspiraling into the ω Cen hole, the waveform is a scale: it weighs whatever hardware shares the inspiral's neighborhood, independent of that hardware's luminosity, temperature, or duty cycle.

The channel's reach must be stated carefully, because it is short. The secular dephasing from an exterior axisymmetric mass falls as $(r_{\rm orb}/r_{\rm torus})^3$ per orbit, so a debris torus or swarm at 10^2–10^3 AU accumulates negligible phase against an inspiral at $\sim 10^{-3}$–10^{-2} AU; we make no claim of debris-torus reach, and the channel does not repair the relic-searchability gap of §3.3. What it constrains is mass at or inside the inspiral track, $r \lesssim 10^{-2}$ AU, and that limitation is well matched to the hypothesis: Paper A's thermodynamic gradient concentrates hardware at exactly those depths, and the envelope of §2 permits it there. A LISA-band inspiral through an occupied deep envelope would traverse the swarm itself. An order-of-magnitude scope for the fiducial system: an inspiral accumulates $\sim 10^5$–10^6 radians of orbital phase over a multi-year observation, and matched filtering resolves phase drifts of order unity, so fractional perturbations at the 10^{-6}–10^{-5} level in the enclosed mass along the track are in principle measurable, a sensitivity to swarm masses far below any other channel's floor. Turning that scope into a forecast requires real waveform modeling, including the degeneracies with the astrophysical environmental effects already cataloged in the literature; we defer that calculation, and any figure it would support, rather than publish an unmodeled curve.

Two properties carry into the adjudication. The channel is two-sided, like R: a measured *vacuum* inspiral, dephasing consistent with zero environmental mass, counts against $H_{\rm eng}$ at the depths the hypothesis most values, and the framework of §4 scores it accordingly. And it inherits the spin channel's schedulability problem: the generic per-IMBH IMRI rate is $\sim 10^{-9}$–$10^{-6}\,{\rm yr}^{-1}$ (Mandel et al., 2008; Amaro-Seoane, 2018; Fragione et al., 2018), with Arca Sedda et al. (2021) the current superseding compilation for dense-cluster rates (LISA detection 0.02–60 $\rm yr^{-1}$, in-cluster IMRI formation probability 5–50 per cent rising with IMBH mass), and ω Cen's own rate is narrower still, $(4\text{–}8) \times 10^{-8}\,{\rm yr}^{-1}$ (§5), so the channel is population-level, a constraint LISA accumulates across the Galactic IMBH census rather than a scheduled ω Cen observation. It enters the information forecast of §5 as the fourth channel on those terms.

3.6 Predicted-signature table

Table 1 summarizes what each channel looks like under the engineered hypothesis and the two astrophysical nulls defined in §4.1.

The table exposes the central inferential difficulty, stated plainly in Paper A and now quantified: silence is predicted by everything. $H_{\rm eng}$ and $H_{\rm q}$ are near-degenerate in every electromagnetic channel while the system is dormant. The channels that separate them are spin (engineered selection favors $a_\star \gtrsim 0.9$; the growth history modeled for ω Cen favors $a_\star \sim 0.05$–0.10 under isotropic capture, §3.3), the R statistic (active only during accretion episodes above the §3.2 threshold), EMRI dephasing (active only if an inspiral occurs; §3.5), and the waste-heat floor (which separates them only near the sensitivity ceiling). This degeneracy structure, rather than any single measurement, is what the adjudication framework must manage. Two channels proposed elsewhere in the technosignature literature, gamma rays from harvested Hawking radiation[1] and gravitational waves from a superradiant Kerr transmitter[2], fall outside the table

[1] Crane and Westmoreland (2009) ask whether starships or power sources can be built on the Hawking radiation of an artificial black hole; their hole is of million-tonne scale and radiates in gamma rays. The channel is real for manufactured small holes and empty for this paper's target: a $2 \times 10^4\,M_\odot$ hole has a Hawking temperature of order 10^{-12} K, about twelve orders of magnitude below the microwave background, so no gamma-ray residue from Hawking output is available to observe.

[2] Jackson and Benford (2020) propose gravitational-wave beacons, one architecture being a superradiant Kerr power station of the black-hole-bomb type driving a transmitter built from small orbiting black holes, with line frequencies from 100 Hz to the GHz band. Superradiant extraction needs only a spinning Kerr hole, so an IMBH is in principle the right object; the published transmitter architecture is not, being assembled from manufactured small holes, and the channel is directed transmission rather

Table 1: Forward-modeled observables under the three hypotheses of §4.1. "Silent" means below current limits (for the mid-infrared point source, the forecast ETC depths of §5). GRMHD baseline from Raha et al. (2026).

Channel	$H_{\rm eng}$ (engineered)	$H_{\rm q}$ (quiescent IMBH)	$H_{\rm sub}$ (remnant subcluster)
MIR point source	floor at $10^{-4}P_{\rm comp}$; may sit near the forecast ETC depths	silent; $L \propto \dot{M}$ of ambient gas only	silent; no compact source
Radio continuum	silent while dormant; self-absorbed if fueled (no R information; §3.2)	silent or fundamental-plane track (Mahida et al., 2026)	silent
Variability (R statistic, X-ray/optical)	$R \ll 1$ if accreting above the §3.2 threshold	$R \sim 1$ if accreting	n/a
EMRI/IMRI dephasing	environmental phase drift if hardware sits at or inside the inspiral track (§3.5)	vacuum inspiral	no inspiral (no massive central object)
LISA EMRI spin	high-tail spin, $a_\star \gtrsim 0.9$ (selection at arrival; §3.3)	low spin, $a_\star \sim 0.05$–0.10, from the minor-merger growth history the ω Cen models find under isotropic capture (González Prieto et al. 2025, Appendix B); the $\chi \simeq 0.7$ attractor applies to comparable-mass histories elsewhere in the class (Fishbach et al. 2017; Gerosa and Berti 2017)	no EMRI, but a comparably dense stellar-mass black-hole population instead produces resolvable mHz binaries (SNR $\sim$ few$\times 10^4$ at 2 mHz over a 4-yr mission), a distinct LISA signature in its own right
Fast-star kinematics	IMBH-like point mass	IMBH point mass	extended mass profile
MSP timing	point mass, 2.20× the merger endpoint if the spin-up ran late (§5)	point mass at the merger endpoint	extended potential (current data lean this way; Bañares-Hernández et al. 2025; Colom i Bernadich et al. 2026)

for reasons of scale and of architecture, respectively.

4 The adjudication framework

4.1 Hypothesis space

We adjudicate among three hypotheses about the ω Cen center:

- $H_{\rm q}$: a quiescent IMBH of mass 10^4–$5 \times 10^4\ M_\odot$, electromagnetically silent for want of fuel, spin drawn from an uninformative prior. The parsimonious reading of the fast stars (Häberle et al., 2024b). Appendix B derives a sharper spin prediction for this hypothesis than "uninformative" ($a_\star = 0.049 \pm 0.021$ at the merger endpoint mass, 0.084 at the fiducial mass), and we keep the uninformative prior here deliberately: it widens $H_{\rm q}$'s spin prediction, which can only reduce the contrast a spin measurement would draw between the two hypotheses, so it is conservative for the null. The cost is zero on 2026 data because the spin channel is unscored, and it is the informative prior that should be used the moment a measurement exists.
- $H_{\rm sub}$: no IMBH; the central mass is an extended subcluster of stellar remnants of $\simeq 2$–$3 \times 10^5\ M_\odot$ within the core, as favored by the pulsar-timing analysis (Bañares-Hernández et al., 2025).
- $H_{\rm eng}$: an IMBH hosting engineered infrastructure within the envelope of §2, with observables forward-modeled in §3 and parameters ($P_{\rm comp}$, $f_{\rm sink}$, R, $a_\star$) assigned the priors stated in Appendix C.

The menu is explicit and extensible; adding hypotheses (an exotic compact object, instrumental systematics on the fast stars, the abandoned-system variant $H_{\rm eng}^{\rm relic}$ of §3.3) changes bookkeeping, never structure. $H_{\rm eng}^{\rm relic}$ is degenerate with $H_{\rm q}$ on every current channel (§3.3) and is therefore not separately scored until a spin measurement exists. We follow the convention that $H_{\rm eng}$ is reported only as odds against the best-performing null, never against a strawman. On the current record the selection is inert: every active channel's null predicts silence with probability one under both $H_{\rm q}$ and $H_{\rm sub}$, so each per-channel Bayes factor takes the same value against either (Appendix C); the selection binds only once a channel activates.

4.2 Per-messenger Bayes factors

For each observational channel i with data d_i, the Bayes factor between hypotheses H_a, H_b is

$$K_i^{ab} = \frac{\int \mathcal{L}(d_i \mid \theta_a)\, \pi(\theta_a)\, d\theta_a}{\int \mathcal{L}(d_i \mid \theta_b)\, \pi(\theta_b)\, d\theta_b}, \tag{8}$$

with channel likelihoods built from the forward models of §3 and, following radio-SETI practice, an explicit interference/systematics component mixed into every likelihood (Sheikh et al., 2021; Li et al., 2026) so that "instrumental artifact" is priced inside each channel rather than adjudicated informally afterward. Priors on $H_{\rm eng}$ parameters are pre-registered (Appendix C) and varied over stated ranges in sensitivity analysis; the framework's outputs are always reported as (Bayes factor, prior-sensitivity band) pairs, never as bare numbers.

One evaluation rule governs the interface with Paper C's campaign, and it matters more than it looks. Paper C promotes a candidate to adjudication when a pre-registered frequentist trigger fires, and that promotion is data-dependent. The likelihoods above are therefore always evaluated on the

than residue, so it enters the framework of §4.1 through the extensible menu rather than as a forward-modeled row of Table 1.

complete campaign record, every epoch and every channel including the ones that recorded nothing, never on the promoted subset alone. The trigger layer allocates follow-up effort and carries no inferential weight. Scoring the selected data as though it were the whole dataset would require the conditional form $K \mid \text{trigger} = K \times P(\text{trigger} \mid H_b)/P(\text{trigger} \mid H_a)$, and at Paper C's global threshold of $p < 5 \times 10^{-7}$ that correction is ~ -14 nats against H_{eng}: large enough to move anything from the candidate band of §4.4 to null-favored. The worked example of §5 follows the full-record rule, scoring six channels of which five are nulls.

One-sided channels have a capacity. Where a channel's data are a non-detection and the null predicts that non-detection with probability one, as the mid-infrared channel does today, the Bayes factor is bounded below by the dormancy prior: $\ln K_i = \ln[f_d + (1 - f_d)P(\text{quiet} \mid \text{active})] \geq \ln f_d$, with equality in the limit of infinite sensitivity. At the fiducial $f_d = 0.5$ the whole channel is worth 0.69 nats however deep the photometry goes, and, absent a detection, it can never produce evidence in favour of H_{eng}. We report each such channel's capacity and spent fraction alongside its value, because the quantity a program should be ranked on is the headroom, not the potential.

4.3 Cross-messenger combination

Channels are combined multiplicatively where independent, $K_{\text{tot}} = \prod_i K_i$, with three corrections. Two are imported from gravitational-wave counterpart methodology (Ashton et al., 2018; Breschi et al., 2024). First, a *coincidence term*: hypotheses that predict correlated anomalies across channels (as H_{eng} does for MIR excess and low R during an accretion episode) earn a likelihood contribution from the observed coincidence structure itself, computed from the joint forward model rather than the product of marginals. Second, a *data-side independence audit*: channels sharing calibrators, atmospheric paths, or reduction pipelines are grouped and their shared systematics marginalized jointly before multiplication.

The third is hypothesis-side and is easy to miss, because it is a property of the hypothesis rather than of the instruments. A dormant installation is quiet in every electromagnetic channel at once, so the dormancy fraction f_d is a single nuisance parameter shared across channels, and the product rule marginalizes it once per channel instead of once in total. Any parameter shared this way must be marginalized jointly:

$$K_{\text{tot}} = \frac{f_d \prod_i P(d_i \mid \text{dormant}) \;+\; (1 - f_d) \displaystyle\int \prod_i P(d_i \mid \text{active}, \theta)\, \pi(\theta)\, d\theta}{\prod_i P(d_i \mid H_b)}. \tag{9}$$

The difference is not cosmetic. With n non-detection channels each returning $P(\text{quiet} \mid \text{active}) = 0.5$ and $f_d = 0.5$, Eq. (9) gives $\ln K = -0.47, -0.63, -0.68$ at $n = 2, 4, 6$, saturating at $\ln f_d = -0.69$ as the capacity argument of §4.2 requires, while the product of marginals gives $-0.58, -1.15, -1.73$ and diverges. The product rule would let a menu grow evidence against H_{eng} without bound by adding degenerate silent channels, which is the mirror image of the mis-scoring that §6 warns about for omitted hypotheses. The correction is inactive in the worked example below, where the second live channel carries no dormancy parameter, and becomes necessary on the day a second one activates.

All three corrections are conservative in effect: the coincidence term rewards only pre-registered correlation patterns, the data-side audit only ever weakens evidence, and the joint marginalization only ever moves $|\ln K|$ down.

4.4 Decision thresholds

We adopt the logarithmic odds scale of Kass and Raftery (1995) with pre-registered action bands, stated here for $\ln K$ of H_{eng} over the best null:

Band	$\ln K$	Action
Null-favored	< 0	routine monitoring; $H_{\rm eng}$ parameter space shrinks
Uninformative	0 to 1	no action; report in campaign updates
Anomaly	1 to 3	targeted follow-up on the discriminating channels
Strong anomaly	3 to 5	independent-team replication; data release
Candidate	> 5 sustained across ≥ 2 messengers	community adjudication per post-detection protocols

These bands are on the Bayes factor, not on posterior odds, and the distinction carries a consequence we state rather than bury. Posterior odds are prior odds times K, and this paper sets no prior on $H_{\rm eng}$: the epistemic-status box of §1 makes everything conditional on the optimization premise, and that premise is the quantity a prior would have to price. The bands are therefore resource-allocation thresholds, specifying what the campaign does next, and they carry no belief content on their own. A reader who wants a belief statement must supply prior odds; at a plausible 10^{-6} per target, $\ln K = 5$ still leaves posterior odds near 1.5×10^{-4}. The same gap governs class-level use: applied across the candidate hosts of Paper A's Table 1, prior odds must scale as $1/N$ or the "candidate" band will be crossed by chance somewhere in the sample. Bayes factors supply that multiplicity protection automatically once the prior odds are stated, which is a further reason to state them.

Two design points. The "candidate" band requires multi-messenger support by construction, honoring Paper C's two-messenger rule; no single channel, at any significance, can reach it, because single-channel anomalies are where every historical false alarm has lived (Sheikh et al., 2021). And the kill conditions of Paper A map onto the same scale in the other direction: the pre-registered falsifiers (e.g., LISA measurement of low spin; resolution of the mass tension in favor of $H_{\rm sub}$) enter as ordinary likelihood terms and drive $\ln K$ negative, so confirmation and falsification run through one pipeline rather than two standards. The framework is deliberately instrument-agnostic: nothing in §4.2–§4.4 references ω Cen, and the machinery applies unchanged to any technosignature target with a defined null menu, complementing the qualitative Rio scale (Forgan et al., 2019) with a quantitative interior.

5 Worked example: Omega Centauri today

We now run the current ω Cen data through the framework. Inputs: the fast-star kinematics (Häberle et al., 2024b), the N-body modeling (González Prieto et al., 2025), the MSP timing bound (Bañares-Hernández et al., 2025) with the TRAPUM 2026 mass limit (Colom i Bernadich et al., 2026), the deep radio silence (Mahida et al., 2026; Tremou et al., 2018), and the JWST infrared limits (Chen et al., 2026). The computation implements the Appendix-B model (deliberately minimal: radius-marginalized hard-threshold MIR likelihood, log-flat priors, 2×10^6 prior draws); code in the paper repository, results in Figure 4.

- **$H_{\rm q}$ vs $H_{\rm sub}$:** the interesting contest, and the framework's finding is that it is currently a stand-off dominated by the kinematics-versus-timing tension: the fast stars pull toward a point mass, the pulsars toward an extended one, and the combined $|\ln K|$ is $\lesssim 1$ with sign depending on how the two datasets' systematics are weighted. As an independent check, the `imbh-constraints` aggregation engine of the series' public toolchain, run on its nine-constraint curated compilation (Table 2, Appendix D), returns an empty jointly-allowed mass window (formal tension verdict: the kinematic floor exceeds the tightest model-dependent ceiling), confirming that no point-mass value satisfies all published constraints at face value. The framework adds discipline, and no verdict, to a tension the community already knows; per series policy the tension is recorded, never collapsed.

- $H_{\rm eng}$ **vs best null:** $\ln K$ sits in the null-favored band, as it should. The only active channel today is waste heat, and, with the continuous per-filter likelihood of Appendix C marginalized over swarm radius and temperature split against the real JWST/MIRI sensitivity curve, it yields $\ln K = -0.47$ at fiducial priors (prior-sensitivity sweep -1.138, -0.474 and -0.079 at $f_d = 0.1$, 0.5 and 0.9): the electromagnetic silence sometimes informally cited as consistent with engineering is, in the accounting, mildly against $H_{\rm eng}$, because the forecast ETC depths exclude 76 per cent of the active-installation prior volume, while the radio, R-statistic, spin, and kinematic channels are inactive or degenerate and contribute nothing. The mass channel is the one exception and is priced below. The radius prior matters: narrowing its upper bound from the fiducial 4×10^3 to 10^3 AU, excluding the coolest outer configurations, raises detectability to $\ln K = -0.52$, confirming that cool, extended swarms are what preserve $H_{\rm eng}$'s viability. The leak prior's lower bound is an engineering estimate (§3.1, Appendix A.3): repeating the computation with the prior extended down to $1 - f_{\rm sink} = 10^{-6}$ gives $\ln K = -0.37$ instead of -0.47, so a softer transport bound softens the conclusion without reversing it. A second sensitivity is the fuel ceiling, and it is more than a sensitivity, because the fiducial choice conditions $H_{\rm eng}$ on a corner of its own parameter space without saying so. The prior extends $P_{\rm comp}$ to $7.6 \times 10^5\ L_\odot$, which is the ceiling at near-extremal spin with the outflow suppression of hot low-Eddington flows itself suppressed: the thin-disc branch of §3.4, in its engineered state. Truncating instead at the ADIOS-suppressed natural ceiling ($\sim 2.6 \times 10^3\ L_\odot$) moves $\ln K$ to -0.37. Rather than pick a corner, we also run the ceiling as a mixture, drawing it per sample from the spin fork ($a_\star = 0.998$ on the thin-disc branch, $a_\star \sim U(0.035, 0.07)$ on the MAD branch, the equilibrium band of §3.4, with $\eta \propto a_\star^2$) and from the suppression state (unsuppressed, or suppressed by a factor log-uniform over 10^2–500), at equal weights on each. That gives $\xi = 0.608$ and $\ln K = -0.363$. The corner weights recover the endpoints exactly: all thin-disc and unsuppressed returns the fiducial -0.474, all MAD and suppressed returns -0.244. Conditioning the ceiling on spin rather than fixing it at the engineered corner is therefore worth about $+0.11$ nats to $H_{\rm eng}$, and the fiducial number is the least favorable of the three to it. A third, and now the largest of the three, is the lower edge of the same prior: $P_{\rm comp}$ is drawn from $1\ L_\odot$ upward, a boundary that coincides numerically with the warm instrument limit and has no physical warrant. Because that boundary has no defensible value, we report the whole curve rather than a second stipulated point. Lowering the floor from $1\ L_\odot$ through 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, and $10^{-6}\ L_\odot$ moves ξ through 0.755, 0.684, 0.614, 0.549, 0.493, 0.410 and $\ln K$ through -0.474, -0.419, -0.366, -0.321, -0.283, -0.229: monotone, roughly 0.05 nats per decade, and never reaching zero, because a floor below the instrument limit adds undetectable prior volume without ever adding detectable volume. Since $\ln K$ here is a ratio of prior volumes, every boundary is a lever and all are now tabulated in Appendix C rather than only the two varied at first (Appendix G). The total is dominated by two structural quantities rather than by observational ones: the present evidence budget is set by the dormancy prior (the data enter only through the excluded active-prior fraction $\xi = 1 - P(\text{quiet} \mid \text{active}) = 0.755$, which is the part the data determine), and the channel is capped at $\ln f_d = -0.69$ in total, of which 68 per cent is spent. The framework's present value is the forecast, not the number. The surviving configurations are dormant or low-power, deep-envelope systems (Figure 2), and they are also post-spin-up. Paper A's Phase 4 drives the hole toward high spin by sustained near-Eddington feeding, which at the fiducial mass is $6.6 \times 10^8\ L_\odot$: four to five orders of magnitude above the concealment ceiling of §3.1, and not concealable by any architecture in the envelope. The episode's duration is set by the mass it must deliver, not by a round number. Everything in this paragraph and the next is a thin-disc-branch statement in the sense of §3.4, and on the MAD branch the growth factor is 1.010–1.020 instead of 2.20 and the consequences below largely vanish. On the thin-disc branch,

reaching Thorne's radiation-limited $a_\star = 0.998$ from $a_\star \approx 0$ requires $M_f/M_i \approx 2.20$ (Bardeen, 1970; Thorne, 1974)[3], and at the Salpeter e-folding time $t_S \approx 4.5 \times 10^7$ yr for $\eta = 0.1$ accretion, that mass gain takes $\ln(2.20)\, t_S \approx 3.5 \times 10^7$ yr. $H_{\rm eng}$ at ω Cen therefore requires that the spin-up episode has already ended, which is a statable constraint rather than a loose reading of "dormant", and it carries a second observable, secular core depletion, that §6 flags as testable against existing surface-density data. It also carries a mass prediction the spin-up requirement makes on its own: the $\approx 2.20\times$ growth factor applies on top of whatever mass the merger history alone delivers, so $H_{\rm eng}$ at ω Cen predicts $M \approx 2.20 \times M_{\rm merger} \approx 2.20 \times 5 \times 10^4 \approx 1.1 \times 10^5\, M_\odot$ against the merger-only $M(H_{\rm q}) \approx 5 \times 10^4\, M_\odot$ (González Prieto et al., 2025), a second discriminant between $H_{\rm eng}$ and $H_{\rm q}$ that does not wait for LISA. The channel is priced here against the TRAPUM 2026 ceiling of $10^5\, M_\odot$ (Colom i Bernadich et al., 2026) (`figs/fE_mass_channel.py`), and the result retires the earlier claim that it was already observable (Appendix G). The two hypotheses inherit the same growth history, so the same merger-mass prior has to describe it under each; using this section's own log-uniform $[10^4, 5 \times 10^4]\, M_\odot$, the late-spin-up branch runs over $[2.2 \times 10^4, 1.1 \times 10^5]\, M_\odot$ and only its top 6.0 per cent clears the ceiling. Reading TRAPUM's 90 per cent limit as a likelihood ratio $\lambda_T = (1 - C)/C = 0.11$ across the boundary, and weighting the late branch at $w = 0.5$, the channel returns $\ln K_{\rm mass} = -0.03$ (band -0.005 to -0.05 over $w = 0.1$–0.9; -0.016 to -0.030 over $C = 0.68$–0.99). The mass channel is therefore a live discriminant in principle and a negligible one on 2026 data, and we say so rather than carrying it as an unscored strength. The figure of $1.1 \times 10^5\, M_\odot$ that earlier drafts quoted assumes the merger endpoint as a point value rather than a range; on that reading the same computation gives $\ln K_{\rm mass} = -0.59$, and the whole weight of the channel sits in the difference between those two treatments of one prior. Per series policy we do not collapse the mass tension in either direction, and the kinematic mass measurement is deliberately not scored here, since doing so would arbitrate $H_{\rm q}$ against $H_{\rm sub}$ as a side effect; only the timing upper limit enters.

- **Information forecast:** channels are ranked by expected $|\Delta \ln K|$ per unit cost against their remaining headroom, not their nominal potential, since the mid-infrared channel has 0.22 nats left in total (§4.2) and further depth cannot buy more. The stated metric is an expectation, conditional value times probability of arrival, and the spin channel's two levels of that expectation diverge sharply enough that they need separate rankings rather than one merged list.

  Target-level, ω Cen itself. The ω Cen-specific IMBH–compact-object capture rate from the González Prieto et al. (2025) models is $(4$–$8) \times 10^{-8}\ {\rm yr}^{-1}$, giving $\sim 3 \times 10^{-7}$ probability of ω Cen yielding its own inspiral over a four-year mission. Even against the large conditional value of a decisive spin measurement (a null at $a_\star = 0.049 \pm 0.021$, or 0.084 at the fiducial mass, makes an engineered high spin a many-σ excursion), the expected $|\Delta \ln K|$ this channel delivers for ω Cen itself is $\sim 10^{-6}$ nats, negligible against the other target-level channels. The target-level leaders are therefore (1) resolution of the mass tension by continued MSP timing plus Gaia DR4 astrometry, which arbitrates $H_{\rm q}$ vs $H_{\rm sub}$ and thereby moves $H_{\rm eng}$'s denominator; (2) any detection of accretion at any level, which activates the R statistic, the only cheap channel with large discriminating power in both directions; (3) EMRI/IMRI environmental dephasing (§3.5), the only channel constraining engineered mass, which shares the spin channel's LISA-era timing and schedulability and, like

[3]The 2.20 figure (2.2024 to 5 significant figures) is our own numerical integration of the standard prograde-ISCO accretion spin-up relation $d(\ln M)/da_\star = 1/[L_{\rm ISCO}(a_\star)/E_{\rm ISCO}(a_\star) - 2a_\star]$ from $a_\star = 0$ to Thorne's $a_\star = 0.998$ equilibrium, using the standard Bardeen et al. (1972) ISCO energy and angular-momentum formulas; it asymptotes to $\sqrt{6} \approx 2.449$ as $a_\star \to 1$. Bardeen (1970) and Thorne (1974) are cited jointly as the standard attribution for, respectively, the accretion spin-up mechanism and the 0.998 radiative-equilibrium limit, not as the source of a closed-form 2.20.

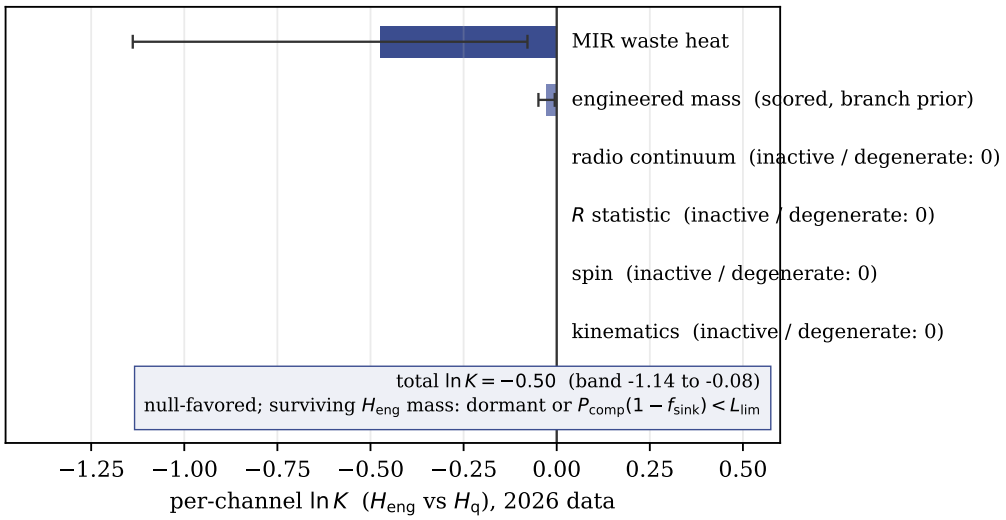


Figure 4: Per-channel $\ln K$ ($H_{\rm eng}$ vs $H_{\rm q}$) at forecast ETC depths, from the Appendix-B computation with the mid-infrared likelihood marginalized over swarm radius. Waste heat carries the evidence budget at -0.474 (whiskers: dormancy-prior sensitivity; extending the leak prior down to 10^{-6} moves the central value to -0.371). The engineered-mass channel is scored rather than assumed inactive, and returns -0.027 at the merger-mass prior this section uses for both hypotheses (whiskers: branch prior $w = 0.1$–0.9); it is a live discriminant carrying almost no present weight, and it reaches -0.59 only if the merger endpoint is read as a point value. The total is -0.501. The empty bars are the framework's forecastable headroom: each activates with the observations ranked in the information forecast.

R, moves evidence in both directions, since a vacuum inspiral counts against $H_{\rm eng}$ at the depths the hypothesis most values; spin itself ranks last at the target level, its 3×10^{-7} arrival probability attached rather than left implicit.

Class-level, the LISA IMBH population. Here spin is the leading channel, because the constraint accumulates across every ω Cen-like system LISA catches rather than waiting on this one target, and probability of arrival is ≈ 1 for the population even though it is $\sim 10^{-7}$ for any single target. The class-level version must carry the spin mixture, since comparable-mass growth histories elsewhere populate the $\chi \simeq 0.7$ attractor, and it inherits the isotropy caveat of §3.3 (Appendix B): the per-target inference only sharpens the population inference if the alignment fraction is itself characterized. The relic analysis of §3.3 raises the stakes on this channel: spin is the one fossil that survives abandonment, so the LISA-era population measurement adjudicates the engineered-history hypotheses for the whole Galactic IMBH population at once, and the per-target frameworks of this paper compose naturally into that population-level test.

Deeper radio silence, by contrast, now moves $\ln K$ weakly for every pairing: for this hypothesis menu the existing limits are already deep enough that further depth buys little discrimination.

6 Discussion

6.1 Limitations

The feasibility envelope depends on cluster-center parameters that carry real uncertainty (mass and number fraction of the remnant tail; the bound-cusp population inside the influence radius) and on a material-strength scale taken from present technology; both are varied in Appendix A. The largest lever is the remnant tail, and within it the perturber mass rather than the number fraction: for the unbound comparison bracket, the $10 \rightarrow 31\,M_\odot$ correction is worth a factor 8.0 in $\langle m_\star^2 \rangle$, against a factor ~ 10 across the whole 0.1–3 per cent fraction bracket; for the bound channel that now sets the operative floor, jointly anchoring $f_{\rm BH}$ and $m_{\rm BH}$ on the extended dark mass narrows the equivalent bracket to a factor 3.1 in kick variance (Appendix A). The cusp existence question is closed by the González Prieto et al. (2025) realizations, and the modeling gap it left behind, that the Monte Carlo modeled only the decoupled unbound flyby channel while the channel that survives, perturbation by bound cusp members, was treated analytically, is now closed: the bound-member diffusion channel and its secular cadence are both Monte Carlo outputs (§2.4, Appendix A). What remains a modeling limitation rather than a parameter is the quasi-static cusp assumption, perturber positions are drawn once per history although the cusp itself relaxes on the same 0.03–0.6 Gyr segregation timescale that builds it, which biases toward longer lifetimes than a resampled population would give; the remaining parameters move the boundaries by factors of a few. The MAD-regulation channel inherits the systematic uncertainties of the GRMHD baseline literature, which is simulation-calibrated rather than observationally calibrated at IMBH masses. The adjudication framework shares the standard vulnerability of Bayesian model selection to unmodeled hypotheses: it ranks the menu it is given, and a surprise outside the menu (an astrophysical mechanism not yet imagined) would be mis-scored until added. The mitigation is procedural rather than mathematical: the menu is public, extensible, and versioned. That procedural claim is no longer unvalidated: Appendix F scores the machinery itself, not the ω Cen models, against three historical cases with an independently known resolution, including a positive control (the 1967 pulsar discovery) that the machinery registers at $\ln K = +7.9$ before correctly collapsing to -0.5 once the rotating-neutron-star hypothesis enters the menu. Two mundane cases (Project Hephaistos, Boyajian's star) are checked for false positives and pass; one data-driven collapse test falls short of its pass threshold as constructed, reported rather than tuned away, and detailed in Appendix F.

Several inputs to this paper come from the companion papers, which are preprints and not yet refereed: the in-cluster fuel census and Phase-4 spin-up episode from Paper A, the 3× crowding factor and the $p < 5 \times 10^{-7}$ trigger from Paper C, and the central density and $4 \times 10^4\,M_\odot$ hole used in the giant-count sensitivity from Paper F. The numbers that depend on them are provisional, and a reader who declines the companions should treat those numbers as conditional.

Two numbers used above deserve explicit qualification at the claim level. **The transport floor.** The geometric content of the bound $1 - f_{\rm sink} \gtrsim 10^{-4}$ is now a calculation (capture cone, beaming gain, étendue, and the plasma-cutoff carrier condition; Appendix A.3); what remains estimated is the re-interception term, the covering-fraction floor on power re-thermalized within the swarm, which a full radiative-transfer treatment of the swarm interior would replace. Its influence on the adjudication is bounded and signed: extending the leak prior down to 10^{-6} moves $\ln K$ from -0.474 to -0.371, well inside the dormancy-prior band of $[-1.138, -0.079]$, so no conclusion of §5 turns on the exact value. **Core depletion is a weaker signature than the series has been claiming.** Paper A and Paper C both list secular core depletion as a Phase-4 observable, and neither had quantified it. Doing so changes its standing, on the thin-disc branch of §3.4; on the MAD branch the delivered mass is $\sim 2 \times 10^2$–$4 \times 10^2\,M_\odot$ and there is no signature to discuss. Spinning the fiducial hole up to Thorne's radiation-limited $a_\star = 0.998$ (§5; $M_f/M_i \approx 2.20$)

requires a mass gain of $\Delta M = 1.20 \times 2 \times 10^4 = 2.4 \times 10^4\ M_\odot$, and the delivered stellar mass is larger still, (2.7–3.4) $\times 10^4\ M_\odot$ once 10–30 per cent radiated and jetted loss is counted. The enclosed-mass model has to be stated, because no single one covers the range: inside the core radius we use the adopted flat core ($\rho_0 = 3 \times 10^3\ M_\odot\,\mathrm{pc}^{-3}$ inside $r_c = 3.6$ pc), and beyond it a Plummer sphere of the same r_c carrying the total cluster mass $3.55 \times 10^6\ M_\odot$, since the flat core extrapolated to 10 pc is not a model of anything. On the flat core the delivered mass is 191 per cent of the mass now inside 1 pc, 24 per cent inside 2 pc, and 4.1 per cent inside the core radius; on the Plummer profile it is 0.91 per cent inside 10 pc. Earlier drafts quoted 1.0 per cent at 10 pc (Appendix G) without saying that the number came from a different model than the three beside it; the flat core extrapolated there would give 0.19 per cent. The fractional deficit is therefore set entirely by the delivery radius, and Paper A's own fuel census puts it at ~ 10 pc, where a sub-per-cent deficit sits below what a cluster surface-density profile can isolate from mass-function and distance systematics. Only fuel drawn from within ~ 2 pc would leave a deficit large enough to see, and that reservoir does not hold enough stars to supply the phase.

The removed stellar mass is not lost from the cluster: it becomes central point mass, so the *total* enclosed-mass profile outside the feeding region is preserved and only the luminous-to-dark split changes, which is the quantity the mass-tension programme of §5 already measures. The endpoint is not degenerate with ordinary growth: $2 \times 10^4 \times 2.2 \approx 4.4 \times 10^4\ M_\odot$ is below the González Prieto et al. (2025) merger-only mass of $\sim 5 \times 10^4\ M_\odot$, so a spin-up episode starting from this fiducial hole does not reproduce the mass the models reach without one, and the final mass carries information about which history produced it. Applied instead to the merger-only endpoint itself as the pre-spin-up mass, the same $\approx 2.20\times$ factor gives the $M \approx 1.1 \times 10^5\ M_\odot$ prediction of §5; the two calculations bracket the endpoint mass depending on when in the growth history the spin-up episode runs. The late-spin-up branch reaches $M \approx 1.1 \times 10^5\ M_\odot$ only if the merger endpoint is taken as a point value; carried over the same $[10^4, 5 \times 10^4]\ M_\odot$ prior both hypotheses share, the branch spans $[2.2 \times 10^4, 1.1 \times 10^5]\ M_\odot$ and only its top 6 per cent clears the TRAPUM 2026 ceiling. The fiducial branch, $4.4 \times 10^4\ M_\odot$, sits below the ceiling throughout. §5 scores the resulting channel at $\ln K_{\mathrm{mass}} = -0.027$: a live discriminant, and on 2026 data a very weak one. What remains of the depletion signature is a lookback limit worth stating on its own: a region depleted at the ~ 10 pc delivery radius refills on the half-mass relaxation time, ~ 12 Gyr ($\log t_{rh} = 10.09$; Harris 1996), and the episode itself lasts only a small fraction of that (3.5×10^7 yr, under one per cent; §5), so the signature persists for a time of order the cluster age after the episode ends. The shorter ~ 4 Gyr core clock ($\log t_{rc} = 9.60$) applies only to a deficit inside ~ 2 pc. Combined with the post-spin-up requirement of §5, the observable and the requirement are in tension only for fuel drawn from that inner region, where a system old enough to be quiet is old enough to have erased its own depletion.

The drainage timescale. The $\sim 10^3$ yr figure of §3.3 is an estimate. The asymmetry it rests on is robust, since relic dust persists around a star while debris around a hole is fed toward the loss cone and removed, but the rate is not. Closing that gap requires a loss-cone refill calculation for the debris population, tracking collisional periapsis diffusion and the coupling of the finest grindings to the ambient flow; we state the gap rather than supply a number that calculation has not yet produced.

6.2 Relation to the series and beyond

Within the series, this paper closes the constructive quadrant: A argued the destination is attractive, and §2 finds the destination is habitable by infrastructure, with the thermodynamically preferred region also the dynamically safest. It also arms Paper C's campaign with the scoring machinery its two-messenger policy presupposed, and returns to Paper D a refined survival input (p_s now decomposable into transit and residence terms, the latter bounded here). One dependency flag on that export: the residence-survival figure Paper D imports (≈ 0.8 over 10^8 yr) was derived from the impulsive diffusion floor. It is now

anchored on the bound-remnant channel of Appendix A, the operative one because unbound flybys never bind: 75 per cent of histories survive to 10^8 yr at the envelope edge and 97 per cent at 400 AU, so the ≈ 0.8 figure is bracketed by 0.75 at the edge and 0.97 at 400 AU. Paper D should cite it against that channel. The adiabatically corrected unbound-flyby curve, whose envelope minimum is 7.6×10^8 yr, describes the non-operative channel, and the impulsive floor remains the conservative alternative. Beyond the series, the two exportable products are the envelope method, applicable to any proposed megastructure environment with a stated perturber population, and the adjudication framework, applicable to any technosignature program willing to pre-register its nulls.

6.3 Conclusion

The engineered-IMBH hypothesis survives its first engineering audit and its first quantitative adjudication, in both cases by narrowing: infrastructure persists only within $\sim 4 \times 10^3$ AU of the hole, and the surviving parameter space after the forecast ETC depths is dormant, low-power, and deep. Every forthcoming measurement listed in §5 shrinks it further or breaks it open. Either outcome is progress that a hypothesis without an envelope, a residue model, and a scoring rule could not deliver.

A Flyby Monte Carlo: method and parameters

Method (implemented in `figs/fig1_envelope.py`; fixed seed 20260717, offset per configuration so that no curve depends on the order in which the configurations are run). For each of 40 logarithmic radius bins $a \in [6\,r_g, 4 \times 10^3]$ AU (the grid begins at the Schwarzschild ISCO): (1) 2×10^5 encounters are drawn with impact parameter from the gravitationally focused cumulative distribution $Q(b) \propto v_{\rm rel}^2 b^2 + 2GMb$ truncated at $b_{\max} = 30a$ (contributions beyond carry $< 10^{-3}$ of the kick variance), relative speed drawn as $v = 0.3\sigma + x$ with x a Rayleigh variate of scale $\sigma = 21\,\mathrm{km\,s^{-1}}$, i.e., a Rayleigh distribution shifted upward by 0.3σ (mean $1.25\sigma + 0.3\sigma = 1.55\sigma$; the JSON output of Appendix E records the exact parameterization), below the single-population Maxwellian mean of a star's speed relative to an effectively stationary hole, $\sqrt{8/\pi}\,\sigma \approx 1.60\sigma$; since slower encounters deliver larger kicks, the choice is conservative for swarm survival, and mass from the perturber mass function. The baseline function has two stellar components ($0.35\,M_\odot$ with weight 0.7; $0.6\,M_\odot$ white-dwarf tail with weight 0.3); the fiducial function adds segregated remnants, $1.4\,M_\odot$ neutron stars at number fraction 0.02 and $31\,M_\odot$ black holes (§2.4) at number fraction $f_{\rm BH} \in \{0.001, 0.01, 0.03\}$ with the stellar weights renormalized. The total number density is set per branch by the observed core *mass* density, $n_\star = \rho_0/\langle m(f_{\rm BH})\rangle$, rather than held at a fixed $10^4\,\mathrm{pc^{-3}}$: adding heavy remnants at fixed number would raise the implied mass density to 4.8×10^3, 7.5×10^3, and $1.4 \times 10^4\,M_\odot\,\mathrm{pc^{-3}}$ across the $f_{\rm BH}$ grid against an adopted 3×10^3, and the $10^4\,\mathrm{pc^{-3}}$ figure itself assumed $\langle m\rangle = 0.3\,M_\odot$, which is not the mean of the function this Monte Carlo draws from ($0.425\,M_\odot$ with no remnants). Renormalizing lowers $\Gamma \propto n_\star$ by 1.42, 1.58, 2.50, and 4.54 for the baseline and the three fractions, lengthening every lifetime by the same factors. Which convention is right turns on whether the adopted ρ_0 is the total dynamical density (then this one is self-consistent) or a luminous density with the remnant tail sitting on top of it; the fixed-number convention is retained as a switch in `common.py` and its bound-cusp output is reported alongside the primary in `fig1_results.json`, where it reproduces the pre-revision numbers exactly. A $10\,M_\odot$ configuration is retained in the output so that the change of perturber mass is auditable. (2) Each encounter contributes an impulsive eccentricity kick $\delta = 2(m_\star/M)\min[1, (a/b)^2](v_{\rm orb}/v)$, i.e., $\delta = \delta v_{\rm tid}/(2v_{\rm orb})$ with the $O(1)$ geometric coefficient set to unity; that choice sits on the optimistic side by a factor of ~ 2 in the δ^2 timescale, offset by the conservative velocity sampling above and by the neglect of adiabatic suppression, so the net sits within the stated

factor-few accounting. The baseline curves apply no adiabatic suppression; a parallel run applies the Gnedin and Ostriker (1999) kernel $A(x) = (1+x^2)^{-5/2}$ to δ^2 per encounter, with $x = (b/a)(v_{\rm orb}/v)$, and is the physical estimate (§2.4). We use the power-law kernel rather than the $\exp(-x^2)$ form of Spitzer (1987) because the latter over-suppresses relative to N-body results; the power law is the conservative choice. (3) Per history (2×10^4 per bin), survival time is the random-walk first-passage to $e = 0.5$: encounters to threshold $n_\star = e^2/\langle\delta^2\rangle$ with CLT scatter, divided by the total encounter rate $\Gamma(< b_{\rm max})$, capped at 12 Gyr. The bookkeeping is thus: the 2×10^5 encounter draws per bin characterize the per-encounter kick-variance statistics; each of the 2×10^4 histories then draws its encounter count analytically from that first-passage distribution rather than re-simulating individual encounters, so a history at the median spans $\sim n_\star \sim 10^4$–10^6 encounters depending on bin. Single-passage disruption is impossible over the part of the grid where the impulsive kernel is meaningful: at $a \gtrsim 1$ AU the saturated kick is $\delta \lesssim 5 \times 10^{-3}$ for stellar perturbers and $\lesssim 0.4$ for the rare 31 $M_\odot$ black-hole perturbers (0.13 for the retired 10 $M_\odot$ run). Inside ~ 0.2 AU the saturated form returns $\delta > 1$ for a heavy penetrating passage ($\delta = 4.0$ at 10^{-2} AU; 1.3 for 10 $M_\odot$), and the first-passage step count $n_\star = e^2/\langle\delta^2\rangle$ falls below 10^2; neither the small-kick premise nor the random-walk premise holds there, which is a further reason the impulsive curves are a floor rather than an estimate in the deep envelope, and is why the adiabatic run is carried alongside them. Direct star–element scattering requires approach within $\sim 2Gm_\star/(v_{\rm rel}v_{\rm orb})$ and contributes negligibly at any realistic covering fraction. This unbound-field calculation uses a core-average perturber density (10^4 pc^{-3}) with gravitational focusing capturing only the unbound flux; it is the curve retained in Figure 1(a) as the measure of that channel's adiabatic decoupling, not as an operative floor.

Bound-cusp channel (panel b). Implemented in the same script; species density profiles and joint anchoring in `common.py`. The relaxed Bahcall–Wolf cusp that the González Prieto et al. (2025) realizations form raises the local density of *bound* perturbers, four species (main-sequence and white-dwarf stars at $\gamma_\star = 1.3$; neutron stars, 1.4 $M_\odot$, interpolated $\gamma_{\rm NS} = 1.5$; black holes, $\gamma_{\rm BH} = 2.0$), each following $N_s(< r) = [4\pi n_{s,0} r_{\rm infl}^3/(3-\gamma_s)](r/r_{\rm infl})^{3-\gamma_s}$ normalized against the core-average density $n_{s,0} = f_s N_{\star,\rm core}$ at $r_{\rm infl}$. The black-hole number fraction is anchored jointly rather than assembled from separate sources: $f_{\rm BH,global} = (M_{\rm dark}/m_{\rm BH})/N_{\star,\rm cluster}$ with $N_{\star,\rm cluster} = 7 \times 10^6$, and $f_{\rm BH,central} = 10 \times f_{\rm BH,global}$ with the $\times 10$ segregation enhancement adopted as a stipulated normalization, not derived here. Varying the enhancement over 3–30 moves the joint-anchored central fraction over 0.35–3.5 per cent and the envelope-edge presence probability over 20–90 per cent, with the conditional diffusion floor spanning 1×10^{10} to 2×10^7 yr (`figs/fE_calc7_seg_sweep.json`); the stipulated factor sets where in that range the headlines sit, and the qualitative conclusion (bound remnants set the operative clock; flybys never bind) holds at every setting. The González Prieto et al. (2025) models publish cusp slopes for the black-hole population ($\gamma_{\rm BH} \simeq 2.0$) but no central number fraction, so the factor remains a stipulation anchored on the $r_{\rm infl}$ matching convention rather than a derived quantity; the slopes imply the contrast grows inward as $(r/r_{\rm infl})^{-0.7}$, a factor 5 at the stripping radius. At the fiducial $M_{\rm dark} = 2.5 \times 10^5\, M_\odot$, $m_{\rm BH} = 31\, M_\odot$: $f_{\rm BH,central} = 1.15$ per cent. Because kick variance scales as $n_{\rm BH}\langle m_{\rm BH}^2\rangle \propto M_{\rm dark} m_{\rm BH}$, a lighter perturber is linearly, not quadratically, less damaging at fixed dark mass; the bracket $\{M_{\rm dark} = 2.0 \times 10^5\, M_\odot, m_{\rm BH} = 15\, M_\odot\}$ to $\{3.0 \times 10^5\, M_\odot, 31\, M_\odot\}$ spans a factor 3.1 in kick variance, against the factor 30 the unbound comparison bracket spans. Since $\gamma_{\rm BH} = 2$ gives a shell count linear in radius, the number of bound black holes inside the stripping radius has expectation $\lesssim 1$ everywhere in the envelope; each history draws its perturber count per species as an independent Poisson variate from the expected shell count $\lambda_s(a) = N_s(< \sqrt{2}\,a) - N_s(< a/\sqrt{2})$ rather than using a smooth rate, which is what makes the resulting lifetime distribution bimodal rather than unimodal. Relative velocities are drawn as Maxwellian with the local Keplerian dispersion $s_{{\rm rel},s} = \sqrt{2GM/[(1+\gamma_s)a]}$ (species-dependent, since $x \sim 1$ for bound members and σ does not apply); impact parameters are uniform in b^2 out to $b_{\rm max} = 3a$, set by where

the adiabatic kernel's contribution to the kick variance drops below 1 per cent; gravitational focusing is dropped (it uses the perturber's own mass for an already-orbiting body and is a $\sim 10^{-3}$ correction at $b = a$). The same Gnedin–Ostriker kernel applies, since $x \sim 1$ rather than $\ll 1$ for these perturbers; an un-kerneled run is retained in the JSON (`bound_fid_nokernel`) as a diagnostic only. Survival is the same random-walk first-passage criterion as the unbound channel, with the total rate summed over the four species' Poisson-drawn counts for that history.

Secular cadence. $t_{\rm sec}(a) = (M/m_{\rm pert})(r_p/a)^3 P_{\rm orb}(a)$ for the outermost bound black hole, fixed at $r_p = 4 \times 10^3$ AU (where $N_{\rm BH}(< r) = 1$); quenched by relativistic apsidal precession, $t_{\rm GR}(a) = [a/(3r_g)]P_{\rm orb}(a)$, and by mass precession from the light cusp population enclosed within a (drawn per history from the same $N_s(< a)$ profile, excluding the perturbing black hole), $t_{\rm mass}(a) = [M/M_{\rm enc}(< a)]P_{\rm orb}(a)$; active only where $t_{\rm sec} < \min(t_{\rm GR}, t_{\rm mass})$ and within the isotropic-cusp resonance window, probability 0.775 for mutual inclination in $(39.2°, 140.8°)$.

Results. For the stars-plus-WDs baseline, median impulsive lifetimes are 4.8×10^9, 4.7×10^9, and 3.6×10^9 yr at $a = 10, 10^2, 10^3$ AU: flat, by the scale cancellation $\langle \delta^2 \rangle \propto a^{-1}$, $\Gamma \propto a$ discussed in §2.4. Adding the remnant components lowers the floor in proportion to the added $\langle m_\star^2 \rangle$: medians at the same radii are $(1.6, 1.5, 0.92)\times 10^9$ yr at $f_{\rm BH} = 0.001$, $(1.7, 1.6, 1.3)\times 10^8$ yr at the fiducial $f_{\rm BH} = 0.01$, and $(9.8, 9.0, 8.8)\times 10^7$ yr at $f_{\rm BH} = 0.03$, with envelope minima of 1.6×10^8, 6.9×10^7, and 5.2×10^7 yr respectively. The perturber mass dominates the fraction bracket: the same fiducial run at the earlier $10\,M_\odot$ assumption gives $(9.8, 9.2, 6.2)\times 10^8$ yr with an envelope minimum of 3.6×10^8 yr, so the $10 \to 31\,M_\odot$ substitution is worth a factor 8.0 in $\langle m_\star^2 \rangle$ and displaces the floor further than the whole 0.1–3 per cent bracket does at fixed mass. Varying the mean stellar mass by ×2, the density by ×3, and $e_{\rm cross}$ over 0.3–0.7 moves the floor by additional factors of a few. The quoted minima are min-of-noisy-bins statistics; the script reports the spread over five seeds directly, giving ±20–40 per cent (for example the $f_{\rm BH} = 0.01$ minimum spans 6.9–9.3×10^7 yr), so no hard floor is claimed at better than a factor of two.

The adiabatically corrected run, which is the physical estimate for the unbound channel, sits far above all of these: at the fiducial mass function its median reaches the 12-Gyr cap for $a \lesssim 3$ AU and returns 1.1×10^{10}, 9.9×10^9, and 4.3×10^9 yr at $a = 10, 10^2, 10^3$ AU, with an envelope minimum of 7.6×10^8 yr at the outer edge where the suppression is weakest. The adiabatic parameter at $b = a$ runs from 4.1×10^3 at the ISCO to 2.2 at the stripping radius. The operational reading is that the unbound-flyby channel is removed from the problem across the envelope, and that any process that does bind must involve perturbers bound to the hole, which panel (b) now supplies directly rather than as an analytic estimate.

Bound-cusp results. At the envelope edge ($a = 4 \times 10^3$ AU), a bound black hole is drawn in 24.8 per cent of 2×10^4 histories ($p_{\rm BH,present}$; the analytic expectation from $1 - e^{-\lambda_{\rm BH}}$ at the joint-anchored fiducial). The outcome is bimodal by construction, not merely broad, so the conditional and unconditional statistics are reported separately: conditional on presence the diffusion floor has median 4.3×10^7 yr, while the unconditional median is the 12-Gyr cap and 72.3 per cent of histories reach it because no black hole (or a distant one) was drawn. Earlier drafts reported the unconditional median under the conditional label (Appendix G); the two coincided only while $p_{\rm BH,present}$ exceeded one half, which it no longer does. Interior to the edge, presence becomes rare and the conditional floor becomes severe: at $a = 4 \times 10^2$ AU, $p_{\rm BH,present} = 2.8$ per cent with a conditional floor of 1.4×10^6 yr, and 97 per cent of histories survive to 10^8 yr against 75 per cent at the edge; this is the trade the analytic single-perturber estimate could only gesture at, rare but severe inward, common but mild at the edge. The perturber-mass bracket of the joint anchoring moves the conditional edge floor from 1.8×10^8 yr at $\{2.0 \times 10^5\,M_\odot, 15\,M_\odot\}$ to 4.3×10^7 yr at fiducial and 4.3×10^7 yr at $\{3.0 \times 10^5\,M_\odot, 31\,M_\odot\}$, the factor-3.1 variance bracket stated above; the un-kerneled diagnostic run drops the conditional edge floor to 5.9×10^6 yr, confirming the adiabatic suppression matters by a factor of a few for the bound channel too, as expected since $x \sim 1$ rather than

$\ll 1$ there. The secular clock reaches $t_{\rm sec} = t_{\rm GR} \simeq 3.7 \times 10^7$ yr at $a \simeq 4.0 \times 10^2$ AU, its GR-quenched inner edge, and $t_{\rm sec} \simeq 1.2 \times 10^7$ yr at $a \simeq 9 \times 10^2$ AU against $t_{\rm GR} \simeq 2.6 \times 10^8$ yr there, unquenched; it is drawn in Figure 1(b) only over 4×10^2–4×10^3 AU and is not a survival curve at any point on that range.

A.3: transport bound. The floor $1 - f_{\rm sink} \gtrsim 10^{-4}$ is a calculation in three steps: delivery geometry, carrier physics, and re-interception, with only the last carrying an estimated coefficient.

Delivery. The photon-capture cross-section of the hole is $\sigma = 27\pi r_{\rm g}^2$ (critical impact parameter $3\sqrt{3}\, r_{\rm g}$), so from a platform at $r = 10^2\, r_{\rm g}$ the disposal channel subtends $\Omega_c/4\pi = 6.75 \times 10^{-4}$: isotropic emission delivers less than a thousandth of its power to the horizon, and $f_{\rm sink} \to 1$ requires a beaming gain $\gtrsim 1.5 \times 10^3$. That gain is optically trivial. The capture cone has half-angle $\theta_c \simeq 0.052$ rad, so diffraction demands only an aperture $D \gtrsim 23\lambda$, and the étendue (concentration) limit $1/\sin^2\theta_c \approx 3.7 \times 10^2$ puts the optic area at a few hundred radiator areas. Delivery is not the binding term.

Carriers. The minimum-energy vacuum carrier, wavelength $\sim r_{\rm g}$, energy $hc/r_{\rm g} \simeq 7 \times 10^{-33}$ J, is a ~ 10 Hz wave and cannot propagate: it lies below the plasma frequency of any realistic environment (Paper A, this revision). Inside a fed envelope with $n_e \sim 10^4$–$10^8\,{\rm cm}^{-3}$, $\nu_p \approx 1$–90 MHz, so the minimum propagating carrier energy runs 6×10^{-28} to 6×10^{-26} J, five to seven decades above the vacuum figure. The delivered per-bit disposal cost is therefore set by the plasma cutoff together with multi-bit-per-photon coding at the channel capacity derived in the appendix of Paper A v2.12, which assumes carrier modes kept colder than the 2.7 K sky, and matter carriers, cold mass dropped down the capture cone, are a fallback that evades the cutoff entirely at a higher cost per bit. None of this changes the geometric floor below; it fixes the cost per bit, not the intercepted fraction.

Re-interception. Beamed flux traverses the swarm on its way to the cone and is intercepted with probability of order the swarm covering fraction seen from a typical element, $\gtrsim 10^{-4}$ for the covering fractions that make the swarm computationally worthwhile; the intercepted power is re-thermalized and radiated at swarm temperature. This is the binding term, and its coefficient is the one number a full radiative-transfer treatment of the swarm interior would refine (§6); the sensitivity analysis of Appendix C shows the adjudication does not turn on it.

B Spin channel: random-walk derivation and net-alignment sensitivity

Derivation. For a capture of mass $\mu \ll M$ onto a hole of spin parameter $a_\star = Jc/(GM^2)$, the change in $a_\star$ per event is $\delta a_\star = (\mu/M)\,[\tilde{L}\cos\iota - 2a_\star\tilde{E}]$, with $\tilde{L}$ the specific angular momentum deposited (units GM/c) and ι the inclination of the captured orbit to the existing spin axis. Under isotropic arrival, $\langle\cos\iota\rangle = 0$, the mean drift comes only from the mass term and $a_\star$ is set by the vector random walk of J against a denominator M^2 that grows with each capture. At $a_\star \ll 1$ the walk is free of the Hughes–Blandford damping term, which does not act until $a_\star$ approaches its ~ 0.3 equilibrium, not reached here. Each capture deposits $|\delta J| = \mu\tilde{L}(GM_i/c)$ in a random direction, with $M_i = M_0 + i\mu$; summing in quadrature over $N = (M_f - M_0)/\mu$ events,

$$\langle J^2\rangle = \left(\frac{\tilde{L}\mu G}{c}\right)^2 \sum_i M_i^2 \approx \left(\frac{\tilde{L}\mu G}{c}\right)^2 \frac{M_f^3}{3\mu} \qquad (M_f \gg M_0), \tag{10}$$

so that

$$a_{\star,\rm rms} = \frac{J_{\rm rms}c}{GM_f^2} = \frac{\tilde{L}}{\sqrt{3}}\sqrt{\frac{\mu}{M_f}}, \tag{11}$$

which for a quasi-circular capture from the Schwarzschild ISCO ($\tilde{L} = 2\sqrt{3}$) reduces to $a_{\star,\rm rms} = 2\sqrt{\mu/M_f}$. The M_i^2 weighting means the walk is controlled by the final mass: the last captures are both the largest

angular-momentum depositors and the ones normalized against the largest M^2.

Internal consistency against Mandel (2008)'s scaling. Mandel et al. (2008) gives the general result that IMRI spin-up scales as $\chi \sim m/M$ and rarely exceeds $\chi = 0.3$ for the neutron-star-capture channel; we did not locate a specific stated $123\,M_\odot/a_\star = 0.2 \pm 0.08$ data point in the accessible text to validate against directly, so we check our own formula's behaviour against the scaling instead. The isotropic random walk for a hole gaining half its mass from neutron-star captures, $M_f = 2M_0$, gives $\sum M_i^2 = 0.2917\,M_f^3/\mu$ and $a_{\star,\mathrm{rms}} = 1.871\sqrt{\mu/M_f}$. Setting $a_\star = 0.2$ with $\mu = 1.4\,M_\odot$ gives $M_f = 123\,M_\odot$, an M_f/μ ratio of $\sim$ 88, consistent with Mandel's $\chi \sim m/M$ scaling at order of magnitude and within his stated sub-0.3 ceiling. The spread checks too: a 3-D random walk gives a Maxwellian $|a_\star|$ distribution with $\sigma/\mathrm{mean} = \sqrt{3 - 8/\pi}/(2\sqrt{2/\pi}) = 0.42$, comparable to the ~ 0.4 relative spread generic to this class of random-walk spin-up. The formula's internal behaviour is consistent with the general scaling the literature supplies; we adopt it on that basis rather than importing a specific external number, since Mandel's own worked examples use M_f/M_0 and mass ratios that differ from ω Cen's by more than an order of magnitude each.

Applied to ω Cen. With $\mu = 30$–$40\,M_\odot$ (González Prieto et al., 2025), $M_f = 2 \times 10^4$ (this paper's fiducial mass) to $5 \times 10^4\,M_\odot$ (the González Prieto et al. (2025) endpoint), and $\tilde{L} = 2\sqrt{3}$ (circularized capture) to 4 (direct plunge on a marginally bound orbit):

Corner	$\mu\,(M_\odot)$	$M_f\,(M_\odot)$	$\tilde{L}$	$a_{\star,\mathrm{rms}}$
Minimum	30	5×10^4	$2\sqrt{3}$	0.049
Central	35	5×10^4	$2\sqrt{3}$	0.053
This paper's fiducial mass	35	2×10^4	$2\sqrt{3}$	0.084
Maximum	40	2×10^4	4	0.103

Converting the central rms value to the Maxwellian mean and dispersion gives $a_\star = 0.049 \pm 0.021$ (90th percentile 0.077); the full derived range across all corners is $a_\star \sim 0.05$–0.10. The corners mix two masses and the mass has to travel with the number: the 0.049 and 0.053 entries assume the González Prieto et al. (2025) endpoint $M_f = 5 \times 10^4\,M_\odot$, while this paper's own fiducial $2 \times 10^4\,M_\odot$ gives 0.084 at the same μ and $\tilde{L}$. We quote 0.049 ± 0.021 at the endpoint mass and 0.084 at the fiducial mass throughout, and no longer round either to a single ≈ 0.06, which belonged to neither. This replaces the imported literature value $a_\star \sim 0.1$–0.3 (Mandel et al., 2008), which is the outcome of a different growth factor and mass ratio never derived for this target.

The mass-ratio boundary. No numeric cutoff for the random-walk regime exists in Hughes and Blandford (2003) or its successors; the boundary is not sharp because the underlying physics is continuous in mass ratio. Setting $a_{\star,\mathrm{rms}} = 0.7$, the numerical-relativity attractor value, in the scaling above with $q \equiv \mu/M_f$ gives $2\sqrt{q} = 0.7$, so $q \approx 0.12$: the random walk alone reaches attractor-level spin at $q \approx 0.12$ and produces less below that, as $\sqrt{q}$. Two limits of the construction should be stated with it. The scaling equates a random-walk r.m.s. to an attractor value that numerical relativity obtains from a different mechanism, so $q \approx 0.12$ marks where the walk alone could reach attractor-level spin, not where the attractor takes over. And at $q = 0.12$ the walk runs over $N = M_f/\mu \approx 8$ events, where the $\sqrt{N}$ averaging the derivation rests on is barely established. Fishbach et al. (2017)'s tested range is $q \gtrsim 0.7$, which does not overlap 0.12, so the boundary is not corroborated by that work and we do not claim consistency with it; their $\sqrt{q} \lesssim 0.32$ anti-aligned bound is quoted here only as the one other quantitative q-scaling in the literature. ω Cen sits at $q = \mu/M_f \approx 7 \times 10^{-4}$, a factor ~ 175 below the boundary.

Net-alignment sensitivity. Relaxing $\langle \cos\iota \rangle = 0$ to a net alignment fraction $\epsilon_{\mathrm{rot}} = \langle \cos\iota \rangle$, the coherent

deposit is $J_{\rm coh} = \epsilon_{\rm rot}(\tilde{L}\mu G/c)\sum_i M_i$, with $\sum_i M_i \approx M_f^2/(2\mu)$, giving

$$a_\star^{\rm coh} = \epsilon_{\rm rot}\frac{\tilde{L}}{2} = (1.73\text{–}2.0)\,\epsilon_{\rm rot} \tag{12}$$

for $\tilde{L} = 2\sqrt{3}$ to 4. The mass dependence cancels: unlike the random walk, the coherent channel carries no $\sqrt{N}$ suppression, so it dominates as soon as it exists. It equals the random walk's 0.053 at $\epsilon_{\rm rot} \approx 0.026$–$0.031$, reaches 3× the random walk at $\epsilon_{\rm rot} = 0.08$–$0.09$, crosses the entire imported 0.1–0.3 band between $\epsilon_{\rm rot} \approx 0.05$ and ≈ 0.17 (so at $\epsilon_{\rm rot} = 0.20$ it exceeds the band's upper edge by 15–33 per cent), and reproduces the attractor value by this separate route at $\epsilon_{\rm rot} \approx 0.35$–$0.40$. Above $\epsilon_{\rm rot} \approx 0.45$–$0.5$ the low-spin prediction is gone. ω Cen is a rotating cluster and the inspiraling population is mass-segregated by dynamical friction, a process with no obvious reason to randomize the angular momentum a captured object arrived with, so isotropy is an assumption to test rather than a default. The González Prieto et al. (2025) realizations cannot test it: their code is spherically symmetric and draws transverse directions isotropically, so $\langle\cos\iota\rangle = 0$ is built into them.

C Priors, likelihoods, and computation

Implemented in `figs/fig3_lnk.py` (2×10^6 prior draws; fixed seed). $H_{\rm eng}$ parameters: $P_{\rm comp}$ log-uniform on $[1, P_{\rm fuel}]\,L_\odot$ with $P_{\rm fuel} = 7.6 \times 10^5\,L_\odot$ (the ambient Bondi ceiling of §2.7). That ceiling is the engineered-unsuppressed, near-extremal-spin value and is not cosmetic: it falls $\sim 5\times$ at $a_\star = 0.5$ through the $\eta(a_\star)$ dependence of §2.7, and $\sim 10^2$–$500\times$ under natural outflow suppression, so a fully stated prior is a mixture over ($a_\star$, suppressed/unsuppressed). We instead quote the sensitivity band, since the arithmetic is analytic in $\ln K = \ln[f_d + (1 - f_d)\,P(\text{quiet}\,|\,\text{active})]$: at the fiducial unsuppressed ceiling, $P(\text{quiet}\,|\,\text{active}) = 0.245$ and $\ln K_{\rm MIR} = -0.47$; truncating the ceiling at the ADIOS-suppressed $\sim 2.6 \times 10^3\,L_\odot$ removes much of the detectable prior volume, raising $P(\text{quiet}\,|\,\text{active})$ to 0.377 and moving $\ln K_{\rm MIR}$ to -0.37; the $a_\star = 0.5$ ceiling sits between. Other parameters: $1 - f_{\rm sink}$ log-uniform on $[10^{-4}, 1]$; swarm radius r log-uniform over the envelope $[2 \times 10^{-2}, 4 \times 10^3]$ AU (an uninformative choice within the allowed region; the thermodynamic gradient of Paper A would weight it deeper, which is the conservative direction for $H_{\rm eng}$ since deep swarms are warm and more detectable); dormancy probability $f_d = 0.5$, varied over $[0.1, 0.9]$ for the sensitivity band. The value 0.5 is a stipulated fiducial, not derived from this paper's inputs, so the reported result is the sweep $\ln K = -1.138, -0.474, -0.079$ at $f_d = 0.1, 0.5, 0.9$. A conditional bound exists and is not adopted: if "active" means the spin-up episode alone ($\ln(2.20)\,t_S \approx 3.55 \times 10^7$ yr, §5) and the occupation lasts at least 10^8 yr (§2.6), then $f_d \geq 1 - 3.55 \times 10^7/10^8 = 0.645$ and $\ln K \geq -0.31$. Post-spin-up operation is also active and its duration is not bounded here, so the condition is an assumption and the sweep stands. MIR channel likelihood (v1.0, continuous, OCS-MIRI-1): the waste luminosity $L_{\rm waste} = (1 - f_{\rm sink})P_{\rm comp}$ splits into a warm component $f_{\rm warm}L_{\rm waste}$ and a cool component $(1 - f_{\rm warm})L_{\rm waste}$, each re-radiated as a blackbody from the same swarm radius r via $T_{\rm eff}(L, r)$, giving a two-temperature SED with $f_{\rm warm}$ log-uniform on $[10^{-3}, 0.999]$ (the one new free parameter this build introduces) as the only lever beyond $(P_{\rm comp}, f_{\rm sink}, r)$. Each component's flux density at every MIRI imaging filter's pivot wavelength (F560W through F2550W, the 25.5 μm imager cutoff, Dicken et al. 2024) is compared against the JDox ETC-6.0 point-source sensitivity (STScI, 2026), penalized by the core crowding factor of 3× that Paper C already applies; a configuration is detected iff the summed flux exceeds the crowded limit in *any* filter, combined as a logical OR, never a sum, over the nine filters. This replaces the piecewise step in $T_{\rm eff}$. Filter bandpass width is not sourced and is not folded in; each filter is treated as a delta function at its pivot wavelength, a stated approximation. The crowding factor is inherited from Paper C's NIRCam-calibrated recovery

and has not been verified for the MIRI point-spread function, but the adjudication does not turn on it: removing the penalty altogether ($3\times \to 1\times$) moves $\ln K_{\rm MIR}$ by 0.014 nats, an order of magnitude below every prior boundary in the sensitivity table below. The far-infrared corner (swarm cold enough that its Wien tail is negligible at every MIRI wavelength) is not clamped by an assumed placeholder limit; it falls out of the physics as simply undetected by any filter in the array, which is the conservative state, since no far-infrared instrument is currently operating (OCS-MIRI-DATA). Data = non-detection in all nine filters; $P(\text{quiet}\,|\,\text{active}) = 0.245$, giving $\ln K_{\rm MIR} = \ln[f_d + (1 - f_d) \times 0.245] = -0.47$ at fiducial f_d, band $[-1.14, -0.08]$. A logistic softening of the per-filter threshold, width 0.5 dex in $\log_{10}(F_\nu/F_{\rm lim})$, combined across filters as $P(\text{nondetect}) = \prod_i (1 - p_{{\rm det},i})$, moves $\ln K$ by -0.012 nats, so the threshold shape remains immaterial.

The mid-infrared depth array consumed here is the Exposure-Time-Calculator forecast documented in the sensitivity provenance note (`figs/miri_sensitivity_provenance.md`), not achieved photometry of a recorded ω Cen observation; no such observation exists on the record we cite. The per-filter depths are the ETC-6.0 point-source limits at S/N = 10 over a 10 ks integration under the low-background assumption, so they represent an achievable exposure rather than an arbitrary sensitivity floor; the crowding factor that converts them to the core-confused depths used above is transferred from Paper C's NIRCam-calibrated recovery and has not been independently verified for the MIRI point-spread function. The distinction is one of provenance rather than of number (identical depths yield identical ξ and $\ln K$ under either description), and the worked-example values above are insensitive to depth uncertainty at the levels at issue: varying the depth multiplier over 1–30× moves $\ln K$ between -0.49 and -0.44 (`--crowding-factor`). Whether the channel is labelled a pre-registered forecast or a current-data constraint changes no computation in this paper; it changes only what the campaign may claim about what has already been observed.

Two statistics are reported alongside that number because it is not, on its own, a measure of what the data said. The first is the data-only quantity $\xi \equiv 1 - P(\text{quiet}\,|\,\text{active}) = 0.755$, the fraction of the *active*-installation prior volume the forecast ETC depths exclude; no dormancy prior enters it. The second is the channel capacity: because $P(\text{no detection}\,|\,H_{\rm q}) = 1$, this channel is one-sided and bounded by $\ln K \geq \ln f_d = -0.69$ (§4.2), of which 68 per cent is now spent, against 41 per cent under the hard-threshold model this build replaces. Further mid-infrared depth can buy at most 0.22 nats more, and can never buy evidence for $H_{\rm eng}$.

Prior-edge sensitivity. $\ln K$ here is a ratio of prior volumes, so every prior boundary is a lever and all of them are reported. Each row re-runs the same computation with one boundary moved; each is a command-line switch on the script.

The swarm's covering fraction enters the same machinery as a further command-line switch, `--fcov`: a sparse swarm of covering fraction $f_{\rm cov}$ radiates the same waste luminosity from a smaller area and so runs hotter by $f_{\rm cov}^{-1/4}$ at fixed $L_{\rm waste}$ (§3.1). Two settings:

Boundary moved	Value	ξ	$\ln K_{\rm MIR}$	Switch
— (fiducial)	—	0.755	−0.474	—
Transport floor $1 - f_{\rm sink}$	10^{-6}	0.620	−0.371	`--leak-floor-dex -6`
$P_{\rm comp}$ floor	$10^{-3}\,L_\odot$	0.549	−0.321	`--pcomp-floor-dex -3`
$P_{\rm comp}$ ceiling (ADIOS)	$2.6 \times 10^3\,L_\odot$	0.623	−0.373	`--pcomp-ceiling 2.6e3`
Radius prior	$[0.02, 10^3]$ AU	0.812	−0.521	`--r-range 0.02 1000`
Temperature split floor	$f_{\rm warm} \geq 10^{-1}$	0.747	−0.468	`--split-floor-dex -1`
Threshold shape	logistic, 0.5 dex	0.770	−0.486	`--soft-threshold 0.5`
Covering fraction $f_{\rm cov}$	10^{-1}	0.851	−0.554	`--fcov 0.1`
Covering fraction $f_{\rm cov}$	10^{-2}	0.918	−0.615	`--fcov 0.01`

Figure 5 shows the surviving prior volume behind these rows in the four-parameter active-installation

plane.

The largest lever is the lower edge of the prior. The $P_{\rm comp}$ prior runs upward from 1 $L_\odot$, a boundary that coincides numerically with the warm instrument limit and has no physical warrant; extending it three decades downward is worth +0.153 nats, more than the transport-floor sensitivity (+0.103) and now the fuel-ceiling one as well (+0.101). The temperature-split floor is the weakest lever measured, +0.006 nats, because the total waste luminosity and the swarm radius, not its partition between the two components, set most of the detectability. Stated plainly: the evidence budget is still dominated by the dormancy prior, the data entering only through ξ, but the continuous per-filter likelihood raises ξ from 0.50 to 0.76 against the same forecast ETC depths, because the real JWST/MIRI depths (sub-μJy at the warm filters) exclude far more of the active-installation prior volume than the old order-of-magnitude piecewise limits did; this moves $\ln K_{\rm MIR}$ from -0.29 to -0.47 and the spent capacity fraction from 41 to 68 per cent, without touching the capacity bound $\ln f_d$ itself. The same forward model, a two-temperature swarm SED integrated against the real per-filter MIRI sensitivities, supplies the point-source SED discriminant Paper C's mid-infrared program needs (Program 1, Swanson 2026d): the warm/cool flux ratio across F560W through F2550W is the model's falsifiable prediction for any point-source excess the program detects. Radio, R, spin, and kinematic channels are inactive or degenerate between $H_{\rm eng}$ and $H_{\rm q}$ on current data and contribute $\ln K = 0$ each, as Table 1 requires. The timing channel is no longer among them. §5's spin-up requirement makes $H_{\rm eng}$ predict a different mass from $H_{\rm q}$, so MSP timing does discriminate, and it is now scored at $\ln K_{\rm mass} = -0.027$ (`figs/fE_mass_channel.py`) rather than set to zero by assertion. The magnitude changes no verdict: the channel the paper had been describing as an already-observable discriminant is worth about three hundredths of a nat once it is priced against the same merger-mass prior both hypotheses must share. The $H_{\rm q}$–$H_{\rm sub}$ contest is checked against the `imbh-constraints` v1.0.0 aggregation engine (nine curated constraints; empty joint window, tension verdict), whose mass-window math is anchored by the series' replayed golden and CI parity gate.

D The nine-constraint compilation

The aggregation-engine cross-check of §5 (the `imbh-constraints` library, nine curated constraints, in the OCS toolchain archive v1.0, Zenodo DOI 10.5281/zenodo.20689279) draws on the compilation of Table 2. The jointly allowed window is the span between the strongest lower bound and the weakest upper bound at fiducial parameters; it is empty (8,200 $M_\odot$ against $\approx$ 709 $M_\odot$, tension verdict), the formal statement that no point-mass value satisfies all published constraints at face value. The deep radio limit (Mahida et al., 2026) is not among the nine: it constrains accretion efficiency rather than mass directly.

E Reproducibility

Figure scripts and outputs live beside this manuscript (`paper/figs/`: `common.py` for shared constants, one script per figure, `fig1_results.json` for machine-readable Monte Carlo output). The measurement compilation and aggregation library are part of the public OCS toolchain archive (v1.0, which includes the `imbh-constraints` library), Zenodo DOI 10.5281/zenodo.20689279; interactive calculator versions of the Figure 1 and §2.7 computations are planned for omegacentauri.me. All random draws use fixed seeds; all quoted numbers regenerate from the scripts.

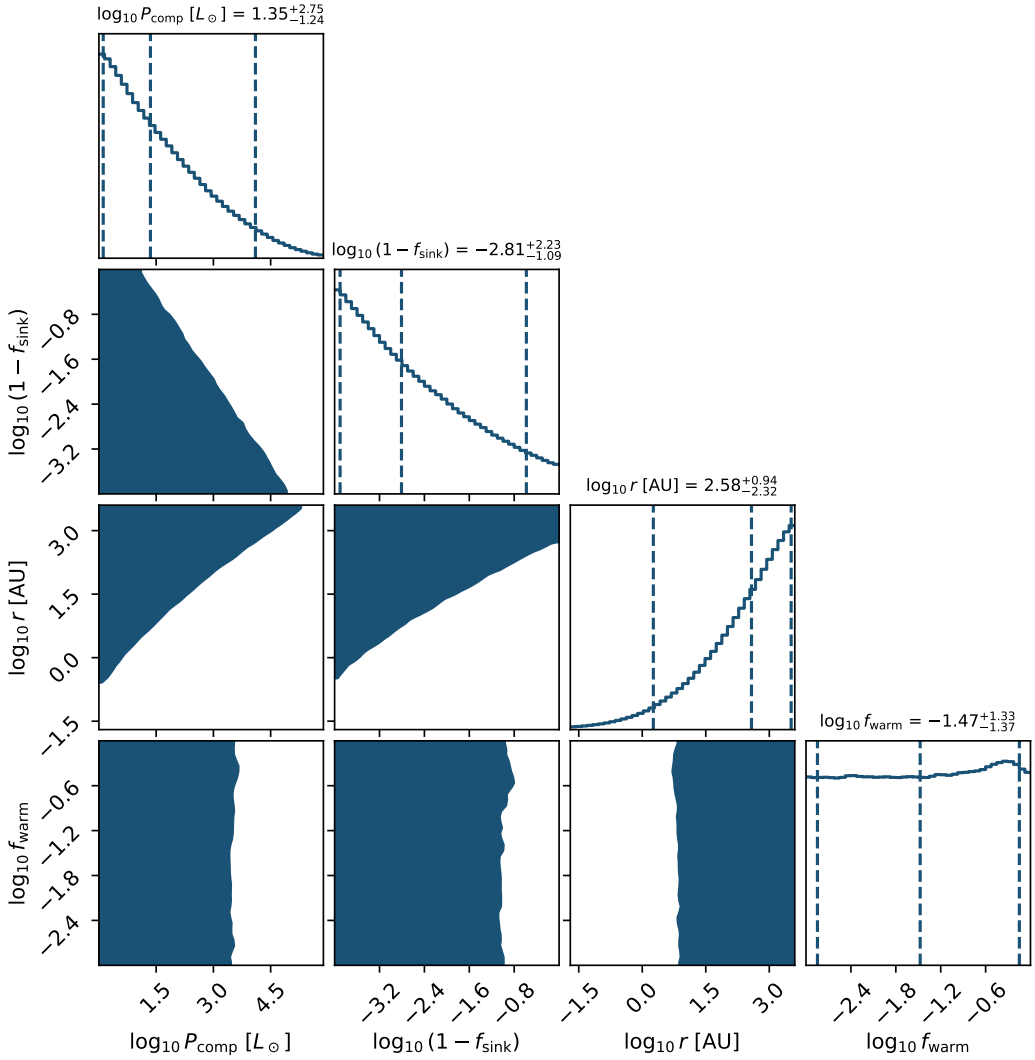


Figure 5: Posterior corner plot of the active-installation parameters under the priors above and the v1.0 continuous MIRI likelihood at the 2026 limits (crowding ×3; the same 2×10^6-draw run as the table). The posterior is the prior times the binary detection likelihood — a configuration contributes iff its two-temperature SED evades every MIRI filter — so the corner displays the surviving prior volume: which parameter combinations the 2026 limits exclude and which survive. The surviving mass sits at low $P_{\rm comp}(1 - f_{\rm sink})$ (the deeply sub-Eddington corner), at large swarm radii (cool SEDs), and is nearly independent of the warm/cool split $f_{\rm warm}$, which the on–off detection structure barely constrains. The excluded 75.5 per cent of the prior volume ($\xi = 0.755$) is the complementary high-luminosity, small-radius region.

Table 2: The nine-constraint mass compilation behind the aggregation-engine cross-check. Windows at face value; "bounds nothing" entries are retained because the engine scores them as detections or context lanes.

Constraint	Observable / method	Window ($M_\odot$)	Source
Noyola et al. 2008	integral-field stellar kinematics (Gemini/GMOS)	detection, $(4\pm1)\times 10^4$	ApJ 676:1008; 10.1086/529002
van der Marel & Anderson 2010	HST proper motions, revised centre	$\leq 1.2\times10^4$ (1σ)	ApJ 710:1063; 10.1088/0004-637X/710/2/1063
Baumgardt 2017	N-body modelling	consistent with zero	MNRAS 464:2174; 10.1093/mnras/stw2488
Häberle et al. 2024	HST proper motions, 7 fast stars	$\geq$ 8,200	Nature 631:285; 10.1038/s41586-024-07511-z
Bañares-Hernández et al. 2025	kinematics + MSP timing, joint	$\leq$ 6,000 (3σ)	A&A 693:A104; 10.1051/0004-6361/202451763
oMEGACat VI 2025	3D kinematic catalogue	context lane (no estimate)	ApJ; 10.3847/1538-4357/adbe67
Chen et al. 2025	JWST NIRCam+MIRI non-detection	$\lesssim$ 709 at fiducial efficiency and density	arXiv:2511.20945; 10.48550/arXiv.2511.20945
Gonzalez Prieto et al. 2025	Monte Carlo N-body growth models	detection, $(5\pm2)\times 10^4$	ApJL; 10.3847/2041-8213/adfd4a
TRAPUM 2026	pulsar timing (MeerKAT+Parkes)	$< 1\times10^5$ (90 per cent)	arXiv:2603.21845; 10.48550/arXiv.2603.21845

F E6 validation of the adjudication machinery

§4's machinery is tested here against three historical cases where the resolution is independently known, per the protocol committed in advance (OCS-E6-PROTOCOL) and implemented in `figs/appendix_d_validation.py`. The object under test is the scoring machinery itself, the explicit-menu requirement, the per-channel Bayes factor, the priced catch-all null, and the action bands, never the ω Cen forward models; what varies per case is the channel likelihood the machinery is fed. LGM-1, the discovery of the first pulsar, is the positive control: the only case that can drive $\ln K$ strongly positive, and the only one that carries a menu-completion test. Project Hephaistos and Boyajian's star are the mundane cases, scored for false positives against a menu that already contained the resolving alternative before the resolving datum arrived. 'Oumuamua is excluded per the protocol: its lightsail proposal has no settled community verdict to calibrate against. Priors and menus for each case were sealed in a dedicated commit containing no scoring code before any $\ln K$ was computed (`paper/e6-priors/priors.json`, commit `f82e11d`); the scoring commit (`96e0d1e`) follows it. Every channel construction is stated in that file and in the script's comments; several are explicit modeling choices rather than re-measurements of the primary data, flagged as such below.

LGM-1. At Epoch 1a (late November 1967, after the fast-recorder capture, before the orbital-Doppler test), the menu is terrestrial interference, artificial signal, and a natural-source catch-all. The catch-all is priced against the fastest coherent celestial periodicity catalogued at that date, of order one hour; CP 1919's coherent 1.337-s period sits 3.4 dex beyond that boundary, and at the fiducial inflation $\lambda = 1$ this channel alone gives $\ln K = +7.90$ against the natural catch-all, satisfying P1 ($\ln K \geq 3$). Its *magnitude* falls in the candidate range, but the candidate band of §4.4 is a two-part rule, $\ln K > 5$ *and* support across at least two messengers, and Epoch 1a is a single channel. LGM-1 therefore registers at strong anomaly by the band rule however large the number gets, which is the behavior §4.4 intends and which earlier drafts contradicted by calling +7.90 "comfortably inside the candidate band" three pages after stating that no single channel can reach it. The framework tolerates catch-all generosity up to $\lambda = 2.63$ before P1 fails; the $(\pi_{\rm nat}, \lambda)$ grid confirms $\ln K$ is invariant in $\pi_{\rm nat}$, as a likelihood ratio must be. Two collapses follow, reported separately per protocol. Collapse B, menu completion: adding the rotating-neutron-star hypothesis to the Epoch-1a menu, priced as though it had existed in November 1967, drives $\ln K$ to -0.50, a 8.40-nat drop that satisfies P2. Collapse A, data with the menu held fixed: adding the null orbital-Doppler test and the discovery of a second independent pulsating source moves $\ln K$ to 6.69 and 4.04 respectively, 2.83 with both; the multiplicity channel is the larger mechanism (-3.86 nats against -1.20 for the Doppler null). This is a finding, not a pass: P3 requires $\ln K$ below 1 after Collapse A and the constructed channels do not reach it, so F3 is reported rather than repaired by re-tuning a prior after seeing the number. The achievable range at Epoch 1a, read off the λ sweep bounds, is $[2.63, 8.00]$, 5.37 nats. That figure is not a passed criterion, because its lower end is set by the top of the λ sweep and the top of the sweep is a choice. Running P6 as a function of $\lambda_{\rm max}$ gives ranges of 0.00, 0.11, 2.74, 4.06, 4.85, 5.37, and 6.43 nats at $\lambda_{\rm max} = 0.5$, 1.0, 1.5, 2.0, 2.5, 3.0, and 5.0: the criterion crosses one nat at $\lambda_{\rm max} \simeq 1.5$ and cannot fail above it. No pre-registered rule fixes $\lambda_{\rm max}$, so P6 is reported as informational rather than passed, and a rule is owed to the protocol.

Project Hephaistos. The dossier's open question, whether Suazo et al. (2024) already discussed background-galaxy contamination at publication or whether Ren et al. (2024) introduced it three weeks later, is resolved by reading the discovery paper's own full text (§3.1, §4): the paper estimates ~ 2 contaminants expected among the seven candidates from Hot DOG chance alignment and states plainly that "the possibility of perfect alignments cannot be ruled out." Background contamination is therefore on the Epoch-1 (2024-05-06) menu by construction, not a later addition, which sets every subsequent trajectory. Scored per candidate for the three with a direct-imaging resolution: G reaches $\ln K = +0.92$

(uninformative) at Epoch 1 and collapses to −3.08 once the off-star VLBI radio position resolves it (2024-05-23), never crossing the strong-anomaly band before that datum existed. D and E follow the same Epoch-1 read, rise to +2.42 (anomaly, not strong anomaly) once the 2026-07-03 archival diagnostics find no contamination signature for them specifically, in contrast to B and C, and collapse to −3.58 once JWST/MIRI resolves both as background galaxies (2026-07-10). All three satisfy P4: no case reaches the strong-anomaly band ($\ln K \geq 3$) before its resolving datum is public, and the anomaly-band excursion for D and E is exactly what that band exists to register.

Boyajian's star. Scope is the short-timescale dipping only, per the dossier's cut; the secular dimming and the contested plate-fading claim are excluded and named as such. The megastructure hypothesis is not on the 2015-09-11 discovery-paper menu; it enters 2015-10-15 (Wright et al.), so the wavelength-dependence channel is inactive at both early epochs and $\ln K = 0$ there, a null result correctly attributed to the channel not yet existing rather than to evidence. The continuous form required by §B is implemented physically rather than as a hard threshold: an anomalous-diffraction extinction efficiency $Q_{\mathrm{ext}}(x)$ is integrated over a log-uniform dust particle-size prior spanning Boyajian et al. (2018)'s own stated constraint region, and compared against the solid-occulter prediction of zero wavelength dependence at any macroscopic scale. At the 2018-01-02 multiband result the achromaticity statistic is read at the 90th percentile of the dust-prior's own predictive, a conservative rather than most-favorable reading of the paper's "at least some dips" language, giving $\ln K = -17.4$ for the artificial hypothesis, decisively null-favored and consistent with the paper's own "inconsistent with... in-line with" language. P4 is satisfied; the channel was never live before the resolving datum. The exact measured reddening slope and its uncertainty were not extracted from the primary photometry in this pass; the scoring instead uses the qualitative constraint the paper itself commits to, a stated limitation rather than a claimed measurement, in the same sense Appendix C's filter-bandpass approximation is stated rather than sourced.

Result. The terminal criteria are P1, P2, P4, and P5, and all four pass; none of F1, F2, or F4 fires, so the framework does not fail validation. Three changes to the criteria themselves are worth stating, because each removed a way the exercise could not have failed. P5 is now a check rather than an assertion: the script reads the committer times of the sealed-priors file and of its own source from git and fails if the seal does not strictly predate the scoring code, where earlier drafts printed a pass unconditionally and described the git workflow in prose. P6 is reported as informational for the reason given above. And every hard-coded likelihood ratio in the three cases is now written as an explicit pair of predictives rather than as a ratio with $P(d \mid H_{\mathrm{art}})$ silently fixed at unity by the numeraire convention; the decomposition reproduces all three of P1, P2, and P3 to the digits quoted here, as it should, since making an assumption visible ought not to move a number. The Boyajian channel keeps one construction that no decomposition repairs: its measurement dispersion is defined so that the megastructure model sits at six standard deviations, so the −17.4 nats it returns restates that choice. Across 2 to 8 standard deviations the channel runs from −1.6 to −31.3 nats for the artificial hypothesis (the script's `ln_r` printout carries the opposite sign, +1.6 to +31.3, because it is the log ratio in favor of the dust alternative), and the verdict is insensitive to the choice only in sign. P3 fails as constructed: Collapse A's data-driven test leaves LGM-1 at $\ln K = 2.83$ rather than below 1, meaning the discriminating data the discoverers themselves treated as decisive (the Doppler null and the second source) carry real but not overwhelming weight in this channel decomposition, a genuinely reachable outcome the protocol was designed to surface rather than to guarantee against. The framework registered LGM-1's anomaly about six months before the community's own resolution (November 1967 detection against the May 1968 Gold paper), tracked Hephaistos candidates G, D, and E through anomaly and back to null without ever crossing strong-anomaly ahead of their resolving data, and correctly returned zero evidence for Boyajian's star before the wavelength-dependence channel existed. The construction choices flagged throughout, most consequentially the one-hour natural-coherence reference boundary for

LGM-1 and the unmeasured Boyajian reddening slope, are stated rather than sourced from a primary catalog or spectrum; a future pass with access to a machine-readable 1967 radio-source catalog and the raw Boyajian et al. (2018) photometry could tighten both without changing the qualitative result.

G Revision record

This appendix lists what changed between versions, so that the body and abstract can state current results only.

v2.8 (2026-10-24). Fact fixes from the Round R10 referee report, adjudicated by the farm (D301, D305, D306) and checked independently. The shear-to-precession comparison is now made at one threshold: precession reaches 2π in 1.7×10^{3} orbits, about 12 days, against 33 orbits for shear, a factor 50 (was a factor 8, which set one radian against 2π). Latency across the swarm is seconds, not milliseconds ($r/c = 9.8$ s at $100\,r_{\rm g}$), and the ISCO period is 9 s. The distance in the scintillation sentence is 5.49 kpc (was 5.4), and the X-ray threshold is 4×10^{-9}–$4\times10^{-8}\,L_{\rm Edd}$ (was 10^{-8}–10^{-7}). The fuel-ceiling lever is +0.101 (was +0.121). The mid-infrared exclusion is attributed to the forecast ETC depths, not to current limits (also in the forward-model table caption and row). The abstract's −2.3 is labelled $\ln f_d$, the channel capacity, not a Bayes factor. The Zenodo record cited for the aggregation library is described as the OCS toolchain archive v1.0 that includes the `imbh-constraints` library, not as a standalone package with an ASCL entry. The envelope edge $0.1\,r_{\rm infl}$ is declared a stipulated prior, with $r_J = 0.81$ pc. The tidal-disruption rate bracket is attributed to Fragione et al. (2018) Eqs. 27 and 8 with an assumed occupation fraction 0.01 (it was attributed to Stone and Metzger), and the spin-up duration quoted from Paper A is 4.2–6.6×10^{7} yr (was 3.4–5.3). The dormancy prior is reported as the sweep −1.138, −0.474, −0.079 at $f_d = 0.1$, 0.5, 0.9, with the bound $f_d \geq 0.645$ given only as a conditional assumption. The worked-example sentence on the joint-marginalisation correction now says the second live channel carries no dormancy parameter.

v2.7 (2026-10-23). Fact fixes from the farm audit D267. The dynamic range between horizon scale and cluster scale is eight orders of magnitude in radius (the paper's own endpoints span 8.3 dex; was nine). The tidal-disruption bracket is 5×10^{-8}–$5\times10^{-7}\,{\rm yr}^{-1}$, the printed cluster-density bracket 5×10^{-6}–5×10^{-5} times the IMBH-hosting fraction 0.01 (was 10^{-8}–10^{-7}); the ω Cen-specific 5×10^{-8} sits at its lower end. Posterior odds at $\ln K = 5$ and prior odds 10^{-6} are $e^{5}\times10^{-6} = 1.5\times10^{-4}$ (was 2×10^{-4}). The composite $\ln K$ at $f_{\rm cov} = 10^{-2}$ is −0.61 in the text (−0.6145, printed as −0.615 in Appendix C; was −0.62). The saturated-kick amplitudes of Appendix A are restated at the adopted 31 $M_\odot$ perturbers: $\delta \lesssim 0.4$ at $a \gtrsim 1$ AU and $\delta = 4.0$ at 10^{-2} AU, with $\delta > 1$ inside ~ 0.2 AU since $\delta \propto a^{-1/2}$ (was 0.14, 1.4 and ~ 1 AU, the retired 10 $M_\odot$ run). The orbital velocities quoted beside the station-keeping budget are ~ 70 km s^{-1} at 4×10^{3} AU and $\sim 4\times10^{3}$ km s^{-1} at 1 AU (was 10^{2}–10^{3}, true only between about 18 and 1800 AU).

v2.6 (2026-10-23). Fact fixes from the farm audits D271 and D253. The van der Marel and Anderson (2010) bound of $1.2\times10^{4}\,M_\odot$ is a 1σ limit (it was labelled 3σ; the primary gives 1.8×10^{4} at 3σ). The cross-references to Paper A are Appendix A of Paper A for the staged architecture and Section 5.2 of Paper C for the wander program. The natural-refilling shortfall is stated against the Paper A requirement of ~ 1–2×10^{-3} stars yr^{-1} (6.8×10^{4} stars over 3.4–5.3×10^{7} yr), i.e. four to five orders of magnitude (was 10^{-4}–10^{-3} and three to four). The MAD-endpoint spin range is described as within 1σ of the natural-capture null (was "at or below"; the null is 0.049 ± 0.021 and the range tops out at 0.07).

v2.5 (2026-10-22). The carrier paragraph of the delivery discussion is aligned with Paper A v2.12: dense coding is bounded by the pure-loss channel capacity, not the Bekenstein bound, and no numerical gain is quoted here.

v2.4 (2026-10-22). The ceiling mixture of §5 was rerun on the MAD equilibrium band $a_\star \sim U(0.035, 0.07)$: the mixture moves from $\xi = 0.649$, $\ln K = -0.393$ to $\xi = 0.608$, $\ln K = -0.363$, the

all-MAD-and-suppressed corner from −0.306 to −0.244, and the total with the mass channel from −0.420 to −0.390; the thin-disc corner is unchanged. The MAD-branch growth factors, episodes and endpoints of §3.4 were reproduced by the script rather than by hand. The statements that $\langle\cos\iota\rangle$ could be read from the González Prieto et al. (2025) realizations were withdrawn: the isotropy of captured orbital planes is an assumption, because that code does not track the orbital plane.

v2.3 (2026-10-20). The abstract was shortened and now opens with the statement that no electromagnetic null can adjudicate $H_{\rm eng}$; the −0.69 cap belongs to the mid-infrared channel and not to the engineered-mass channel. Section 3.4 was rewritten: the fork is the spin-up architecture, extraction at ambient fueling preserves a thin-disc spin (§3.3), the MAD equilibrium is $a_\star = 0.035$–0.07 (Narayan et al. 2022) in place of the earlier 0.1–0.3, and the MAD-branch growth factor, episode, endpoints and fuel ceiling were re-priced by hand (1.0097–1.0198; 9.5×10^2–$3.8\times10^3\,L_\odot$). The $a_\star \sim U(0.1, 0.3)$ ceiling mixture of §5 was computed on the earlier band and its rerun is pending. The residence-survival export was re-anchored on the bound-remnant channel (0.75 at the edge, 0.97 at 400 AU) and the flyby-curve minimum corrected from 3×10^8 to 7.6×10^8 yr. The refill clock for the 10 pc deficit was changed from ∼ 4 to ∼ 12 Gyr. In Appendix F the Boyajian sensitivity range was given its correct sign and −17.9 was reconciled to −17.4. Typographical fixes were made in four places and the sodium-absorption source of n_e was removed.

Earlier versions. (i) Differential precession was first given as the decoherence mechanism for co-radial formations; Keplerian shear decoheres a spread faster (33 orbits against 2.7×10^2) and applies across the whole envelope. (ii) Presence of a bound black hole at the edge was 53 per cent under a fixed-number convention and is 24.8 per cent under the drawn convention of Appendix A. (iii) The giant count inside the influence radius was first taken as $\sim 10^3$, a count of stars there rather than of stars on the giant branch. (iv) The mid-infrared bound began as a single number that assumed the warm case, and the hard-threshold likelihood excluded 50 per cent of the active prior volume against 76 per cent for the continuous per-filter likelihood. Placeholder instrument limits and the "Wien-peak band" and "cross-band leakage" language, which the script never implemented, were retracted. (v) The MAD eruption-power band was first extended a decade lower than the cited cycle supports. (vi) The spin discussion first reduced the discriminating contrast to $\simeq 0.7$ versus $\gtrsim 0.9$ by importing the generic hierarchical-merger caution, and left the isotropy assumption implicit. (vii) The spin-up episode was first given a round duration about three times the computed 3.5×10^7 yr. (viii) The engineered-mass channel was first described as already observable and left unscored, with the kinematic and timing channels at $\ln K = 0$ each; it is priced here at −0.03 nats. (ix) The radius grid first began at $5\,r_{\rm g}$, inside the ISCO, where circular orbits do not exist. (x) Only two prior boundaries were first varied in the sensitivity tables, and the $P_{\rm comp}$ lower edge was not among them. (xi) LGM-1 at +7.90 was first called comfortably inside the candidate band, which contradicted the two-part band rule. (xii) Cusp existence was first left open and carried as a bracket and a second sensitivity lever; the realizations of González Prieto et al. (2025) closed it. (xiii) A $10\,M_\odot$ perturber run printed 2.3 against 0.54 for a velocity-diffusion comparison and was retired at the referee-5 revision. (xiv) The 1.0 per cent deficit at 10 pc came from a different enclosed-mass model than the neighboring figures. (xv) The unconditional median was first reported under the conditional label for the bound-cusp floor, while $p_{\rm BH,present}$ exceeded one half.

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Data availability

No new observational data were generated. All cited measurements are available in the referenced publications. The figure scripts, shared constants, and machine-readable Monte Carlo output described in Appendix E are mirrored publicly at https://omegacentauri.me/papers/source/ and https://omegacentauri.me/papers/figs/, alongside the interactive calculator versions of the same computations.

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